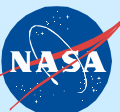


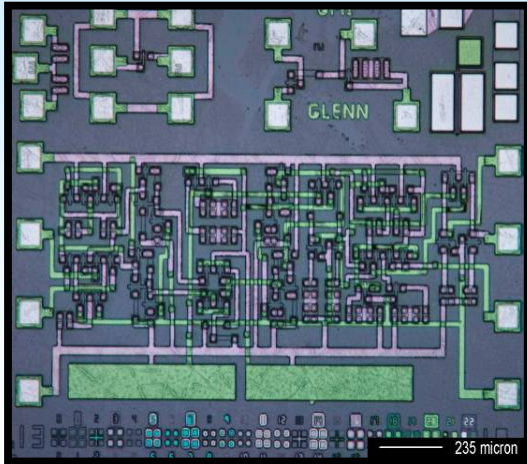
systems in a range of aerospace applications. Descriptions of the sensor elements, their integration into sensors systems, and examples of sensor system applications will be discussed. Finally, suggestions related to the future of sensor technology will be given. It is concluded that smart micro/nano sensor technology can revolutionize aerospace applications, but significant challenges exist in maturing the technology and demonstrating its value in real-life applications.

AEROSPACE SENSOR SYSTEMS: FROM SENSOR DEVELOPMENT TO VEHICLE APPLICATIONS

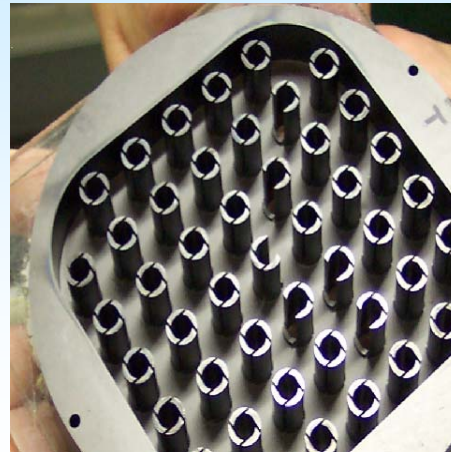
**Gary W. Hunter, Ph.D.
Intelligent Systems Hardware Lead
Sensors And Electronics Branch
NASA Glenn Research Center
Cleveland, OH**



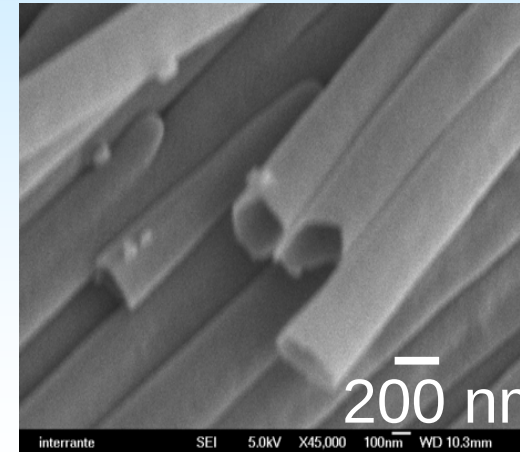
Sensors and Electronics Branch: Scope of Work



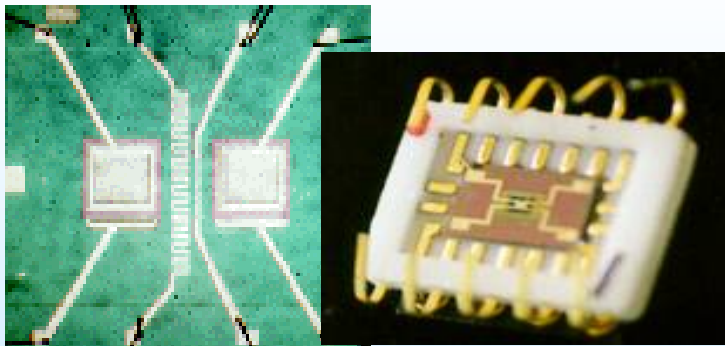
High Temperature SiC Electronics



Micro-electromechanical Systems (MEMS)



Nanotechnology



Chemical Sensors

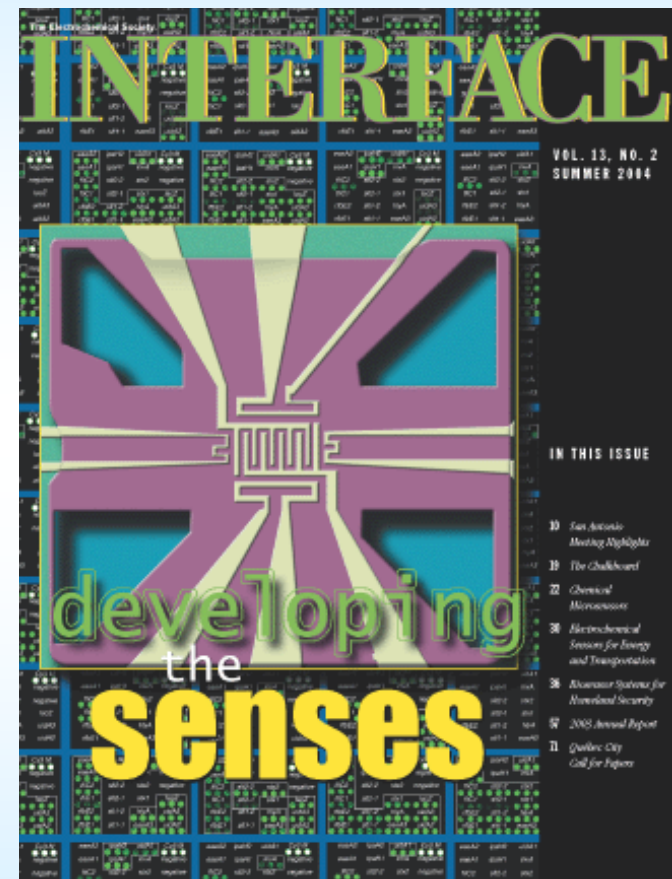


Thin Film Sensors

A RANGE OF SENSOR AND SENSOR SYSTEM DEVELOPMENT



HARSH ENVIRONMENT SENSORS AND ELECTRONICS

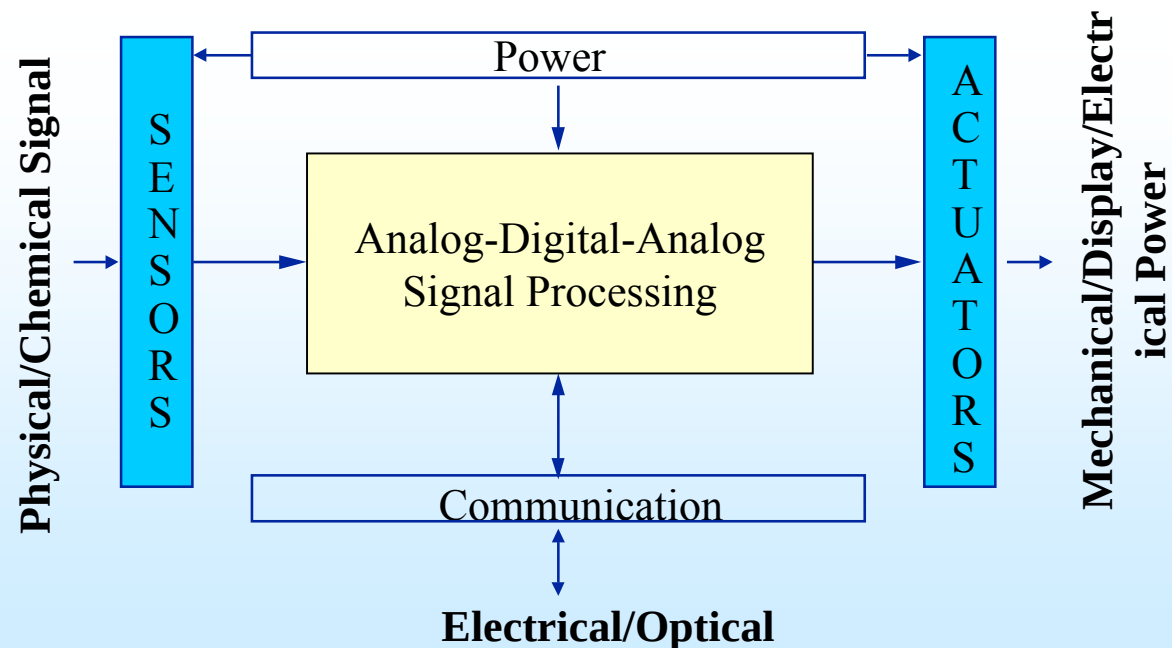


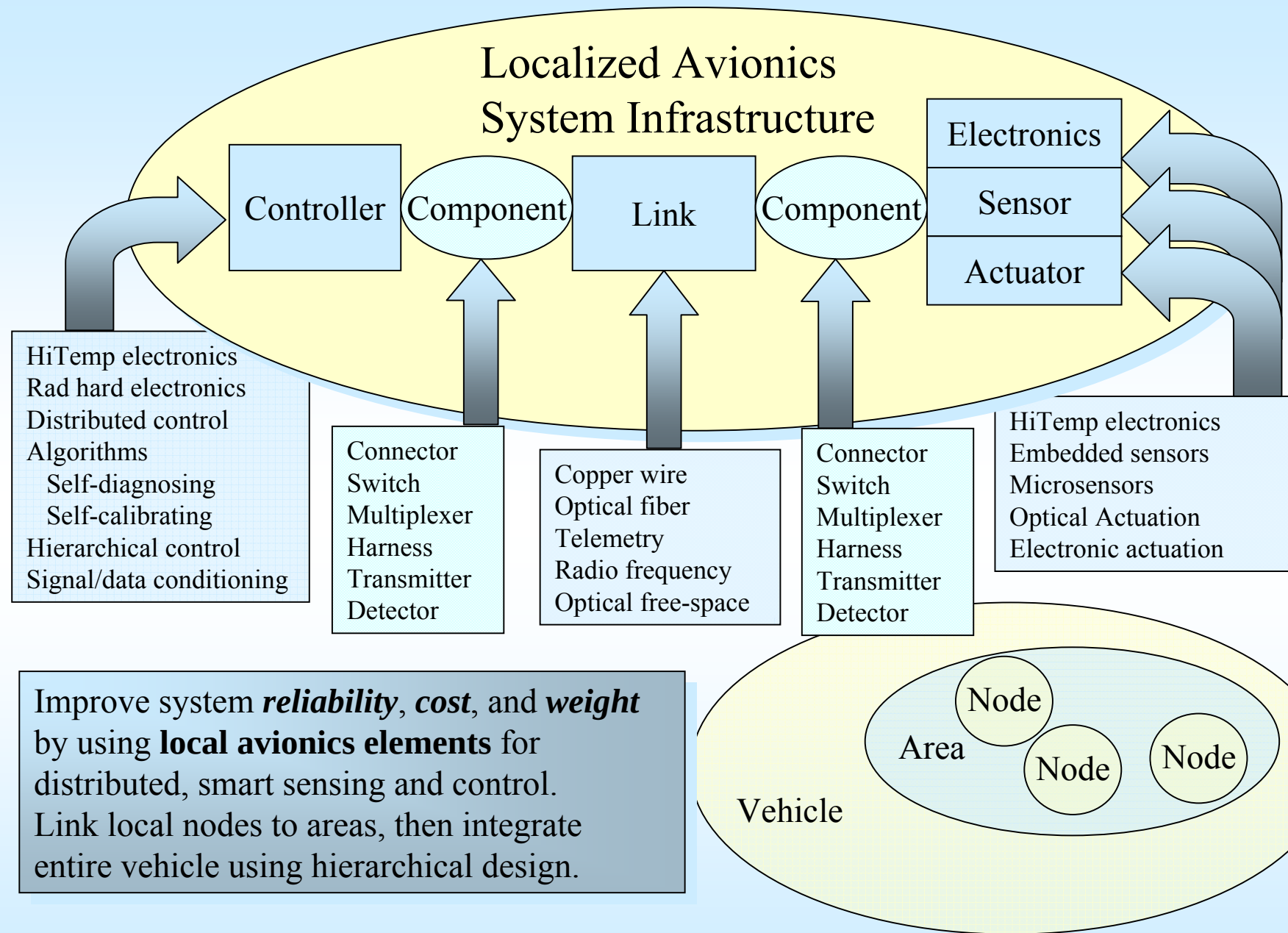
NASA GRC/CWRU O2 MICROSENSOR

MICROSYSTEMS TECHNOLOGY

- THIS PRESENTATION DISCUSSES A RANGE OF GAS SENSOR TECHNOLOGY
- EXAMPLES REVOLVE AROUND MICROSYSTEMS TECHNOLOGY
- EXAMPLES INVOLVE AEROSPACE APPLICATIONS BUT HAVE BROADER IMPLICATIONS
- BASIC APPROACH: DRIVE CAPABILITIES TO THE LOCAL LEVEL/DISTRIBUTED SMART SYSTEMS

Microsystem Block Diagram

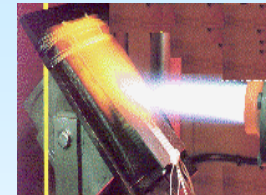




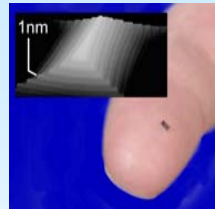
HARSH ENVIRONMENT ELECTRONICS AND SENSORS APPLICATIONS

- **NEEDS:**

- OPERATION IN HARSH ENVIRONMENTS
- RANGE OF PHYSICAL AND CHEMICAL MEASUREMENTS
- INCREASE DURABILITY, DECREASE THERMAL SHIELDING, IMPROVE IN-SITU OPERATION



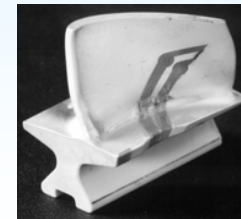
1998 R&D 100 Award



2004 R&D 100 Award

- **RESPONSE: UNIQUE RANGE OF HARSH ENVIRONMENT TECHNOLOGY AND CAPABILITIES**

- STANDARD 500C OPERATION BY MULTIPLE SYSTEMS
- TEMPERATURE, PRESSURE, CHEMICAL SPECIES, WIND AVAILABLE
- HIGH TEMPERATURE ELECTRONICS TO MAKE SMART SYSTEMS



1995 R&D 100 Award



1991 R&D 100 Award

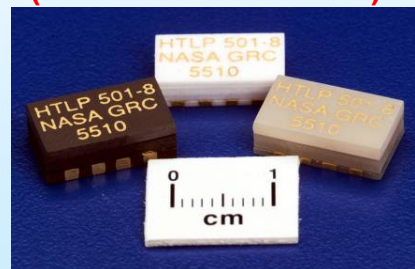
- **ALL-IN-ONE SHOP FOR HARSH ENVIRONMENT SYSTEM APPLICATIONS**

- **ENABLE EXPANDED MISSION PARAMETERS/IN-SITU MEASUREMENTS**

Range of Physical and Chemical Sensors for Harsh Environments



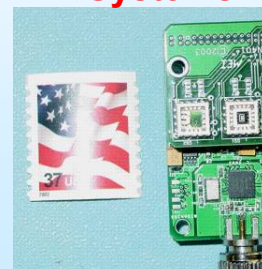
Harsh Environment Packaging
(7000 hours at 500C)



High Temperature Signal Processing and Wireless



Long Term: High Temperature "L and Stick" Systems



BASIC APPROACH: MAKE AN INTELLIGENT SYSTEM FROM SMART COMPONENTS

POSSIBLE STEPS TO REACH INTELLIGENT SYSTEMS

•“LICK AND STICK” TECHNOLOGY (EASE OF APPLICATION)

- **Micro and nano fabrication to enable multipoint inclusion of sensors, actuators, electronics, and communication throughout the vehicle without significantly increasing size, weight, and power consumption. Multifunctional, adaptable technology included.**

•RELIABILITY:

- **Users must be able to believe the data reported by these systems and have trust in the ability of the system to respond to changing situations e.g. decreasing sensors should be viewed as decreasing the available information flow about a vehicle. Inclusion of intelligence more likely to occur if it can be trusted.**

•REDUNDANCY AND CROSS-CORRELATION:

- **If the systems are easy to install, reliable, and do not increase weight/complexity, the application of a large number of them is not problematic allowing redundant systems e.g. sensors, spread throughout the vehicle. These systems will give full-field coverage of the engine parameters but also allow cross-correlation between the systems to improve reliability of sensor data and the vehicle system information.**

•ORTHOGONALITY:

- **Systems should each provide a different piece of information on the vehicle system. Thus, the mixture of different techniques to “see, feel, smell, hear” as well as move can combine to give complete information on the vehicle system as well as the capability to respond to the environment.**

OUTLINE

- **INTRODUCTION**
- **SENSOR ELEMENT**
 - **TECHNOLOGIES AND CHALLENGES**
- **MICROFABRICATED GAS SENSORS**
 - **SENSOR PLATFORMS**
 - **SMART SENSORS SYSTEMS AND DEMONSTRATIONS**
- **SUPPORTING TECHNOLOGIES**
 - **HIGH TEMPERATURE ELECTRONICS EXAMPLE**
- **SYSTEM APPLICATION**
 - **MATURATION FOR ISS APPLICATIONS**
 - **SPACE QUALIFIED ELECTRONICS AND SENSORS**
 - **MOBILE SENSOR PLATFORMS**
- **SENSOR DATA QUALIFICATION**
- **FUTURE DIRECTIONS**
 - **NANOTECHNOLOGY**
- **SUMMARY**

SENSOR ELEMENTS:

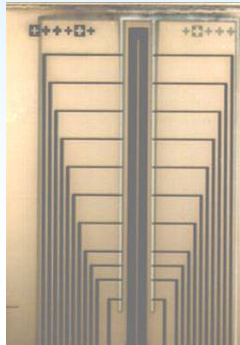
**SPACE APPLICATIONS AND TECHNICAL
CHALLENGES**

TAILOR THE SENSOR TO THE APPLICATION

Thin Film Physical Sensors for High Temperature Applications

Advantages for temperature, strain, heat flux, flow & pressure measurement:

- ◆ Negligible mass & minimally intrusive (microns thick)
- ◆ Applicable to a variety of materials including ceramics
- ◆ Minimal structural disturbance (minimal machining)
- ◆ Intimate sensor to substrate contact & accurate placement
- ◆ High durability compared to exposed wire sensors
- ◆ Capable for operation to very high temperatures ($>1000^{\circ}\text{C}$)



Multifunctional smart sensors being developed

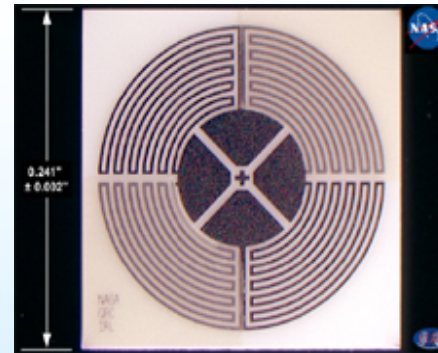
Flow sensor made of high temperature material



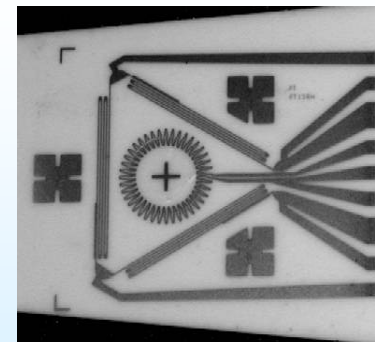
PdCr strain sensor
to $T=1000^{\circ}\text{C}$



Pt- Pt/Rh temperature
sensor to $T=1200^{\circ}\text{C}$



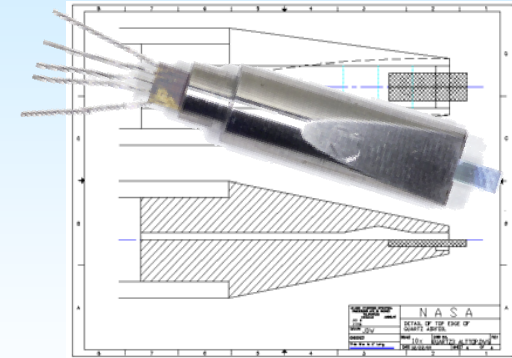
Heat Flux Sensor Array
to $T=1000^{\circ}\text{C}$



Multifunctional
Sensor Array

GRC's Physical Sensor Instrumentation Research History

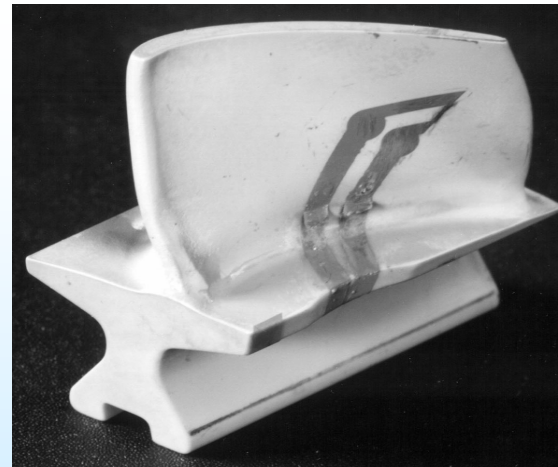
- R&D 100 Awards in 1991, 1995, and 1998
- NASA Group Achievement Award 2003
- Partnerships in Sensor Development:
 - Rolls-Royce, GE Aircraft Engines, Pratt & Whitney
 - Goodyear Tire & Rubber Company
 - University of Rhode Island
 - Air Force Research Lab (NDE Branch)



2003 NASA Group Achievement Award
SiC High Temperature Drag Force
Transducer as part of the Integrated
Instrumentation and Testing Systems pro



1991 R&D 100 Award
PdCr wire strain gauge applied on
Ford Motor Co. exhaust manifold



1995 R&D 100 Award
PdCr thin film gauge applied on Allied-Signal
Engines ceramic turbine blade



1998 R&D 100 Award
Long-lived Convoluted Thermocouple
For Ceramic Temperature Measurement

Novel Thin Film Sensor Technology Development

- Development of Thin Film Sensors with Ceramic, Laminate and Nanostructured Materials
 - Improve techniques for applying high temperature sensors onto complex structures
 - Develop thin film sensors to measure temperature, strain, and heat flux for hot section components
- Technology Challenge: Survivability in Extremely High Temperature Environments ($>1500^{\circ}\text{C}$)
 - Build off of experience fabricating devices on more conventional components
 - Leverage partnerships with University of Rhode Island and NASA GRC Ceramics Branch for ceramic-based materials



Multilayered Ceramic Sensor with Minimal Apparent Strain Sensitivity

Ceramic TC Sputtering Targets fabricated by the NASA GRC Ceramics Branch



Ceramic Thermocouple fabricated at University of Rhode Island

RTD on Fan Blade



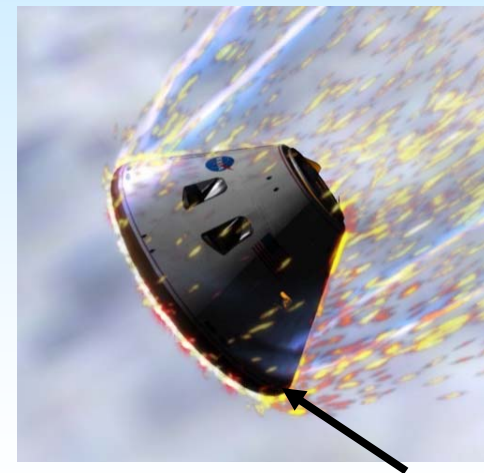
Sputtering System for Thin Film Sensor Fabrication



CEV Interface Seals Instrumentation

Thermal Protection System (TPS) Interface Seals

- Instrumentation for Interface Gap Heating Tests for CEV Heat Shield-to-Backshell Interface Seals
 - Design, fabrication and testing of a packaged miniature heat flux sensor
- Technology Challenge: Sensor Integration with Packaging
 - Leverage RTD Heat Flux Sensor development & Novel Thin Film Sensor effort

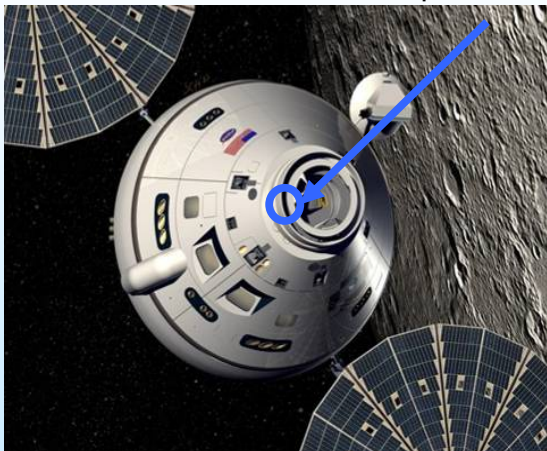


CEV Reentry w/ TPS Seal

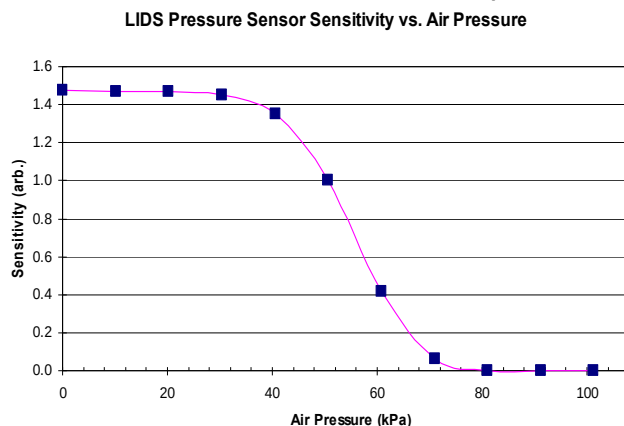
Low Impact Docking System (LIDS) Interface Seals

- Instrumentation to Measure Pressure Between Two Seal Bulbs
 - Concept design, model and lab demonstration of pressure sensor
- Technology Challenge: Sensor Integration with Tight Size Tolerance
 - Miniature size so as to not interfere with the sealing function

LIDS Pressure Sensor Concept Location



LIDS Pressure Sensor Concept Modeled

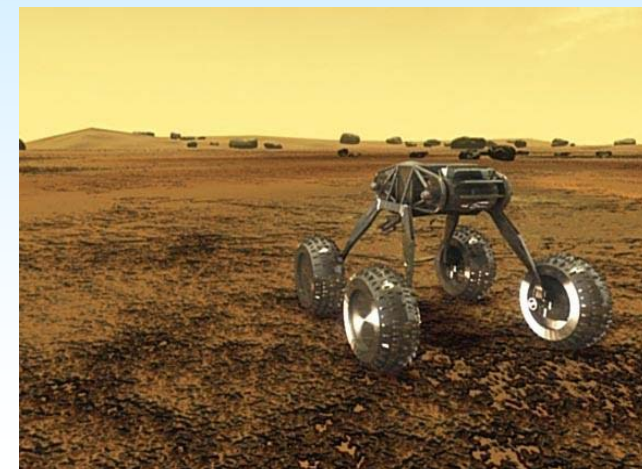


Fabricated Heat Flux Sensor Element

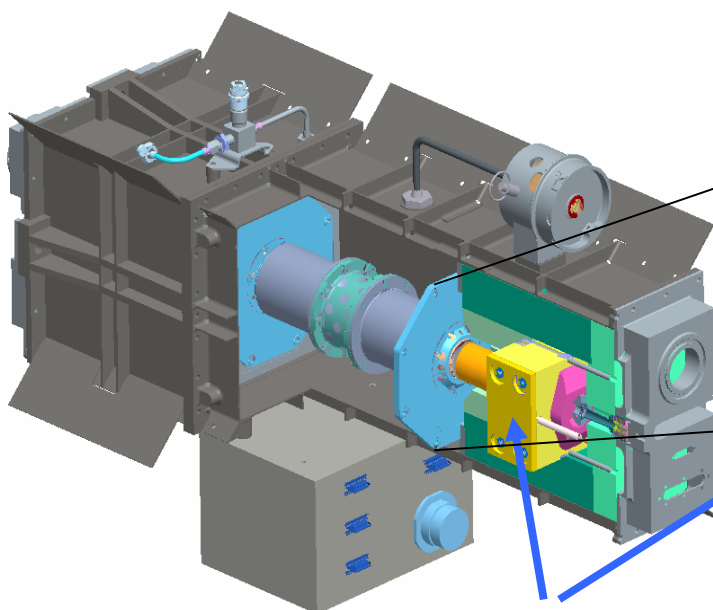


Advanced Stirling Convertor (ASC) Sensors

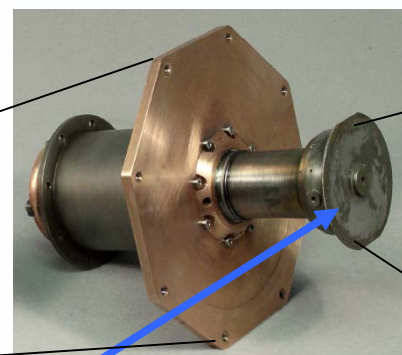
- Development of Heat Flux Sensors for ASC Units for measurement of thermal energy delivered into the converter
 - Allows for a direct measurement of thermal to electrical conversion efficiency in characterizing the ASC units
- Technical Challenge: Sensor Fabrication Methods Compatible with ASC Thermal and Structural Demands
 - Sensor Thermal Conductivity, Strength and Sensitivity satisfied with High Temperature Ceramics



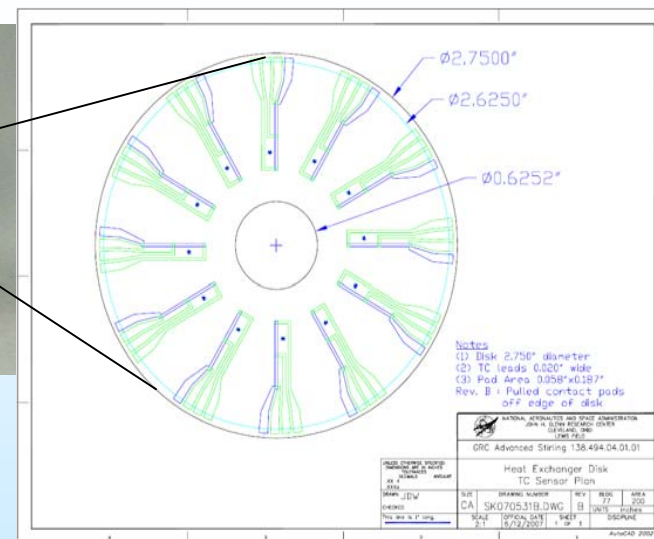
Venus Rover Concept with ASC Cooling



ASRG Engineering Unit with Planned Location of Heat Flux Sensors



ASC-E Converter

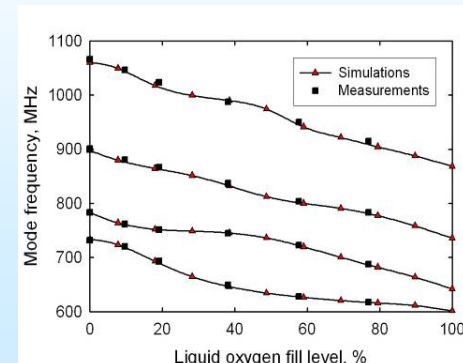
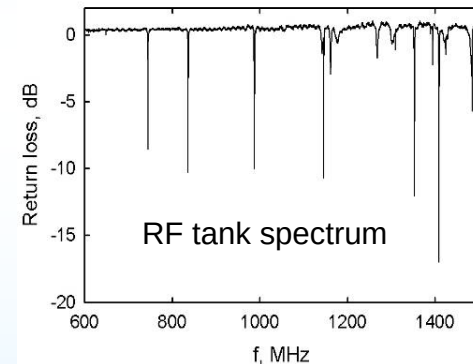
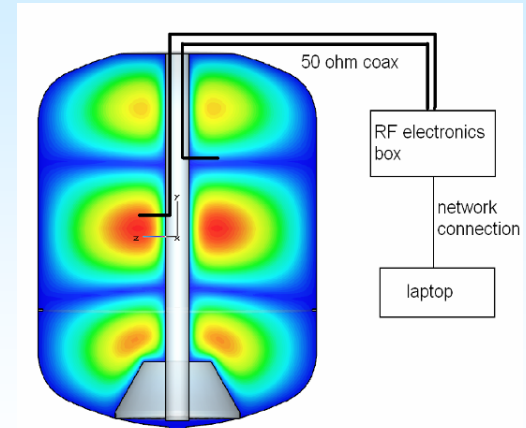


Heat Flux Sensor Design being Fabricated at NASA

Propellant gauging sensor

Radio Frequency Mass Gauge (RFMG)

- Detects electromagnetic modes of a tank using an antenna and RF analyzer
- Mode frequencies are affected in a known way by liquid quantity
- Applicable to dielectric liquids such as liquid hydrogen, oxygen, methane.
- Tests in 58 cu. ft. liquid oxygen tank indicate high accuracy. RMS gauging uncertainty <1% full-scale.
- RFMG is being developed in-house at NASA GRC
- Potential application to EDS and Altair tanks
- Currently at TRL-5



POC: Greg.Zimmerli@nasa.gov

➤ **Future work is designed to push development of the RFMG to TRL 6-7.**

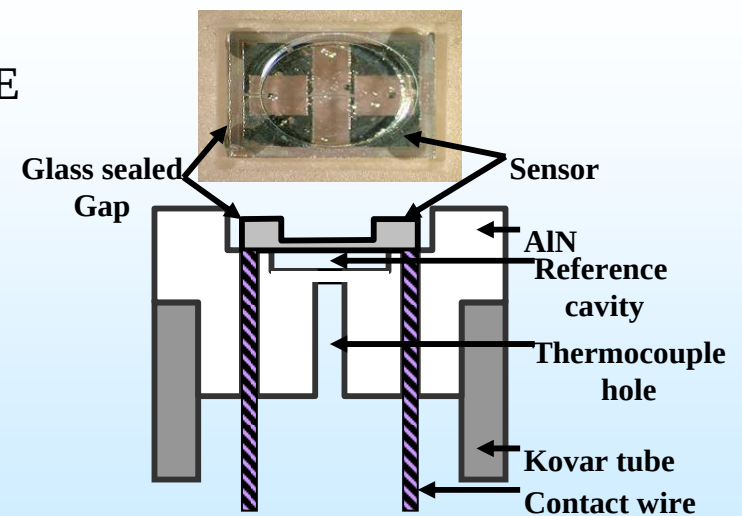
Risk	Mitigation Plan
Limited testing in methane/hydrogen.	Full-scale methane test planned for 2009. Small-scale H ₂ testing planned for 2008.
Unknown RFMG performance in larger scale tanks, and with other internal hardware.	Construct a full-scale mock LH ₂ Lander tank and test RF response with internal hardware (no cryo-fluid).
A high-fidelity flight-prototype instrument unit has never been tested.	Initiate development of a 2 nd generation Engineering Model. Goal is to meet typical thermal, vacuum, vibe, EMI spec's (will try to mimic spec's on Orion CEV).
The structural integrity of the antenna and RF connections has not been determined.	Need to identify/test appropriate connectors. Need to design & vibe-test flight-like antenna.
Performance in low-gravity is uncertain.	1g tilted tank tests. Possible low-gravity aircraft testing. Simulations.

SiC-BASED PRESSURE SENSORS

- **SiC HAS EXCELLENT MECHANICAL PROPERTIES FOR USE AS A HARSH ENVIRONMENT PRESSURE SENSOR ($T > 500\text{ }^{\circ}\text{C}$, SILICON UNDERGOES PLASTIC DEFORMATION)**
- **FORM DIAPHRAM OF SiC AND INTEGRATE WITH ELECTRONICS**
- **WIDE RANGE OF APPLICATIONS**
 - **AERONAUTIC ENGINE APPLICATIONS**
 - **AUTOMOTIVE APPLICATIONS**
 - **WIND TUNNELS**
 - **MATERIAL PROCESSING**
- **ENGINE OPERATION DEMONSTRATED AT 500 C**
- **TECHNICAL CHALLENGES INCLUDE**
 - **SiC DIAPHRHAM FORMATION (ETCHING)**
 - **PACKAGING**
 - **RELIABILITY TESTING**



500 °C SiC pressure sensor

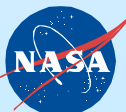


SENSOR PLATFORMS TO SMART SENSOR SYSTEMS: MICROFABRICATED GAS SENSORS

MICROFABRICATED GAS SENSORS



- **COLLABORATIVE EFFORT BETWEEN NASA GRC, CASE WESTERN RESERVE, and OHIO STATE UNIVERSITY**
- **SENSOR DEVELOPMENT RESULTING FROM:**
 - IMPROVEMENTS IN MICROFABRICATION AND MICROMACHINING TECHNOLOGY**
 - NANOMATERIALS**
 - DEVELOPMENT OF SiC-BASED SEMICONDUCTOR TECHNOLOGY**
- **GAS DETECTION IN:**
 - HARSH ENVIRONMENTS**
 - APPLICATIONS BEYOND CAPABILITIES OF COMMERCIAL SENSORS**
- **TECHNOLOGY DEVELOPS PLATFORMS FOR A VARIETY OF MEASUREMENTS**
 - SCHOTTKY DIODE**
 - RESISTANCE BASED**
 - ELECTROCHEMICAL**
- **TARGET DETECTION OF GASES OF FUNDAMENTAL INTEREST**
 - HYDROGEN (H₂)**
 - HYDROCARBONS (C_xH_y)**
 - NITROGEN OXIDES (NO_x) AND CARBON MONOXIDE (CO)**
 - OXYGEN (O₂)**
 - CARBON DIOXIDE (CO₂)**



BASE PLATFORM SENSOR TECHNOLOGY

Integration of Micro Sensor Combinations into Small, Rugged Sensor Suites
Example Applications: AEROSPACE VEHICLE FIRE, FUEL LEAKS, EMISSIONS, ENVIRONMENTAL MONITORING CREW HEALTH, SECURITY

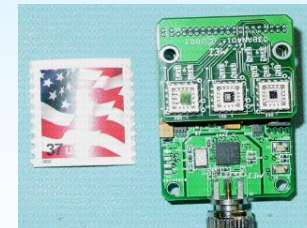
Multi Species Fire Sensors for Aircraft Cargo Bays and Space Applications



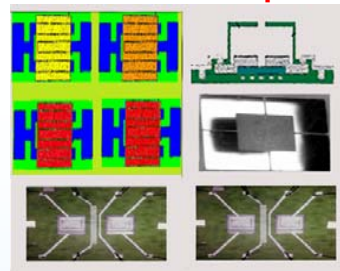
Environmental monitoring (ISS Whitesand Testing)

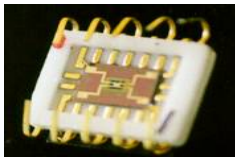


"Lick and Stick" Space Launch Vehicle Leak Sensors with Power and Telemetry



Aircraft Propulsion Exhaust High Temperature Electronic Nose

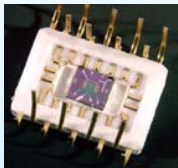




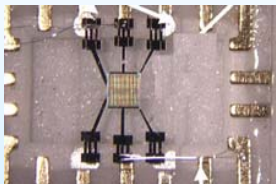
Oxygen Sensor



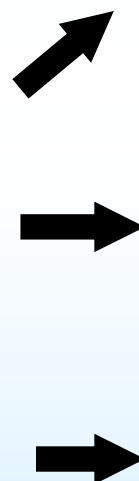
SiC Hydrocarbon Sensor



H2 Sensor



Nanocrystalline Tin Oxide NOx and CO Sensor



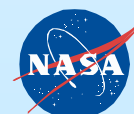
Sensor Equipped Prototype Medical Pulmonary Monitor

Hydrazine EVA Sensors (ppb Level Detection)



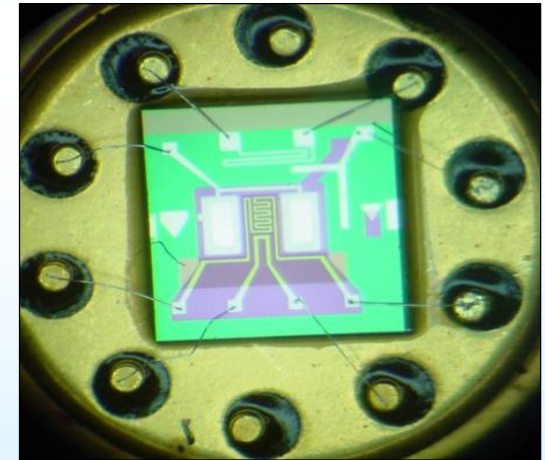
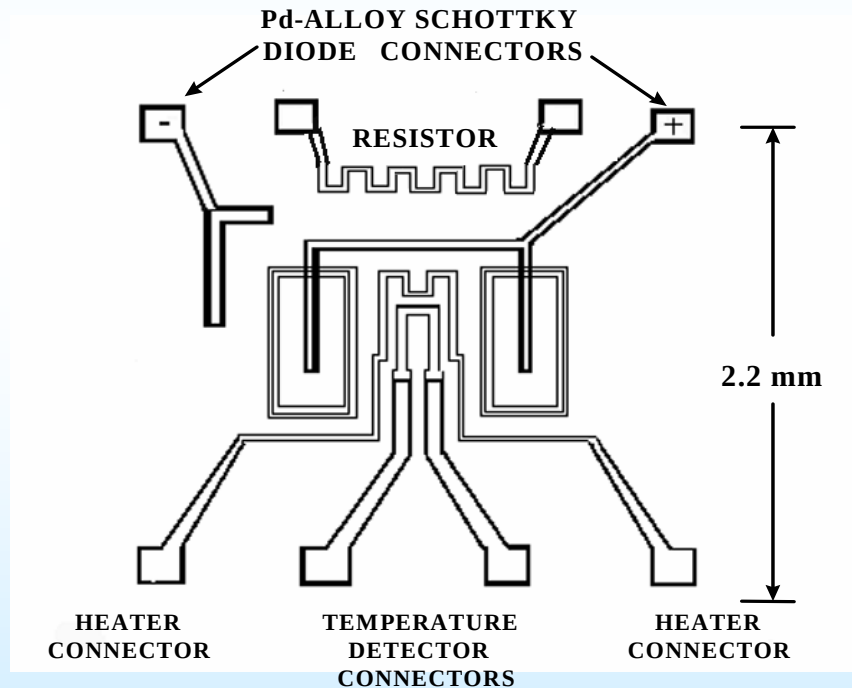
MEI Makel Engineering Inc.

Glenn Research Center



HYDROGEN LEAK SENSOR TECHNOLOGY

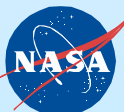
- MICROFABRICATED USING MEMS-BASED TECHNOLOGY FOR MINIMAL SIZE, WEIGHT AND POWER CONSUMPTION
- HIGHLY SENSITIVE IN INERT OR OXYGEN-BEARING ENVIRONMENTS, WID CONCENTRATION RANGE DETECTION
- TWO SENSOR SYSTEM FOR FULL RANGE DETECTION: FROM PPM LEVEL TO 100%



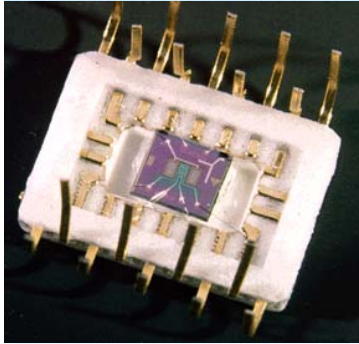
MEI Makel Engineering Inc.



Glenn Research Center



HYDROGEN LEAK SENSOR TECHNOLOGY



- STATUS: OPERATIONAL SYSTEM ON ISS WITH ASSOCIATED HARDWARE
- BEING PREPARED FOR CLV IMPLEMENTATION

1995 R&D 100 AWARD WINNER

NASA 2003 TURNING GOALS INTO REALITY SAFETY AWARD

Shuttle



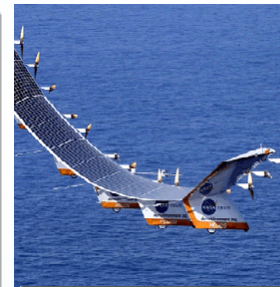
X33



X43



Helios



ISS



Aft Compartment
Hydrogen Monitoring



Hydrogen Safety
Monitoring



Hydrogen Safety
Monitoring



Fuel Cell Safety and
Process Monitoring



Life Support Process
and Safety Monitoring

MEI Makel Engineering Inc.



Glenn Research Center



SIC-BASED GAS SENSOR DEVELOPMENT



- THE USE OF SiC SEMICONDUCTORS ALLOWS SENSOR OPERATION AT TEMPERATURES WHICH ALLOW THE DETECTION OF HYDROCARBONS AND NO_x
- INERT MATERIAL OPERATIONAL IN HIGH TEMPERATURE, CORROSIVE ENVIRONMENTS
- SCHOTTKY DIODE DESIGN FOR HIGH SENSISTIVITY
- TEMPERATURE DETECTOR AND HEATER INCLUDED/OPERATION AT A RANGE OF TEMPERATURES
- WIDE RANGE OF APPLICATIONS
 - EMISSION MONITORING
 - ENGINE HEALTH MONITORING
 - ACTIVE COMBUSTION CONTROL
 - HYDROCARBON FUEL LEAK DETECTION
 - FIRE SAFETY
- PROTOTYPE SENSOR PACKAGE FABRICATED/TESTED IN ENGINE ENVIRONMENTS
- APPROACHES
 - ALLOY ON SiC SUBSTRATE
 - REACTIVE INSULATOR APPROACH
 - BARRIER LAYER
 - ATOMICALLY FLAT SiC

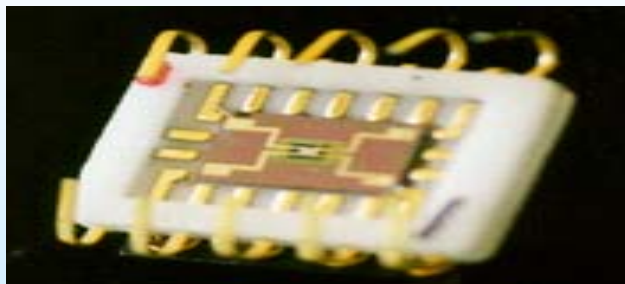
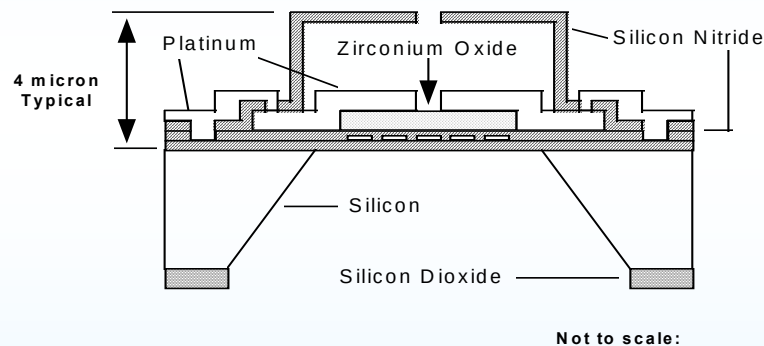
PACKAGED SiC-BASED SENSOR



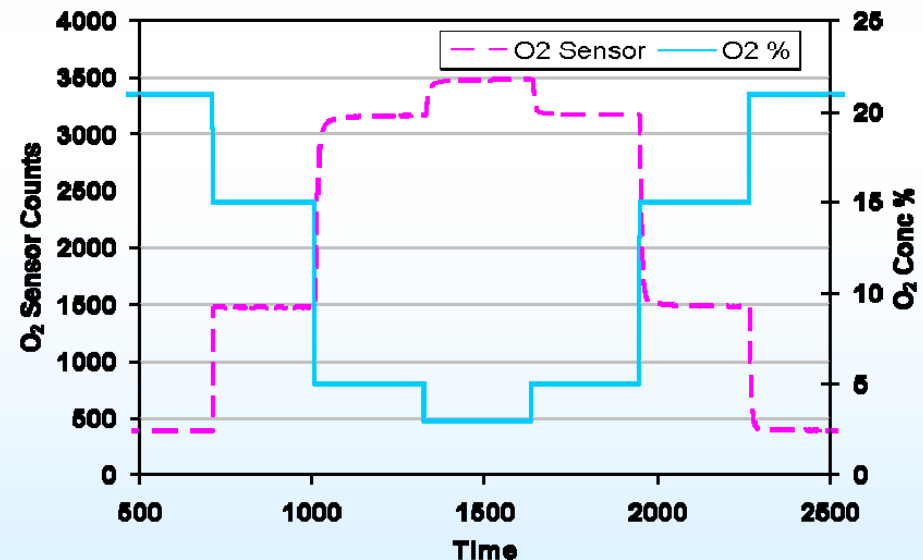
MICROFABRICATED OXYGEN SENSOR TECHNOLOGY



- MICROFABRICATED AND MICROMACHINED FOR MINIMAL SIZE, WEIGHT AND POWER CONSUMPTION (NEAR 100 mW FOR ~500 C OPERATION)
- AMPEROMETRIC OPERATION ALLOWS MEASUREMENT OF OXYGEN OVER A WIDE CONCENTRATION RANGE (0-100%)
- CHAMBER STRUCTURE CONTROLS OXYGEN DIFFUSION RATE
- RELATIVELY MATURE TECHNOLOGY/PACKAGING COULD BE IMPROVED TO DECREASE POWER CONSUMPTION



ZrO₂ Oxygen Sensor

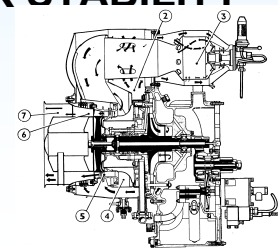
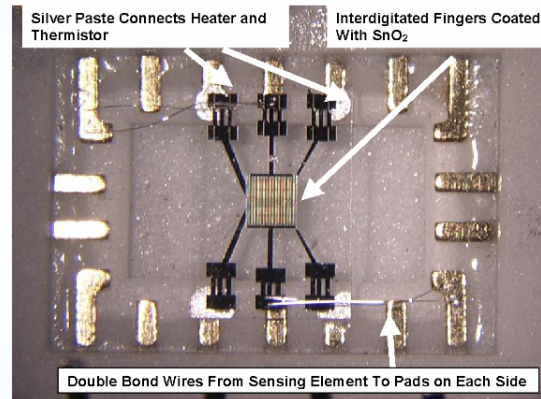
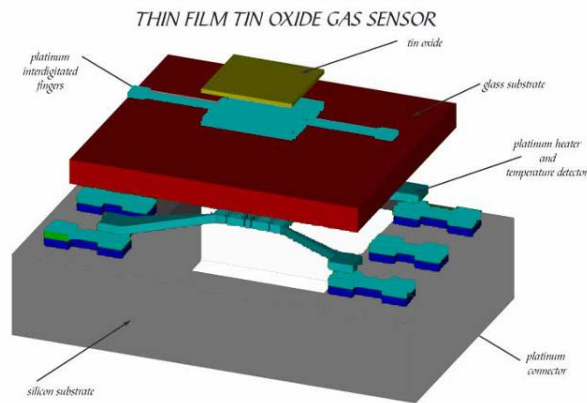


Zirconia Based Oxygen Microsensor Response To Various Oxygen Concentrations

MICROFABRICATED TIN OXIDE BASED NO_x AND CO SENSOR TECHNOLOGY



- MICROFABRICATED FOR MINIMAL SIZE, WEIGHT AND POWER CONSUMPTION
- MICROMACHINED TO MINIMIZE POWER CONSUMPTION AND IMPROVE RESPONSE TIME
- TEMPERATURE DETECTOR AND HEATER INCORPORATED INTO SENSOR STRUCTURE
- NANOFABRICATION OF TIN-OXIDE TO INCREASE SENSOR STABILITY

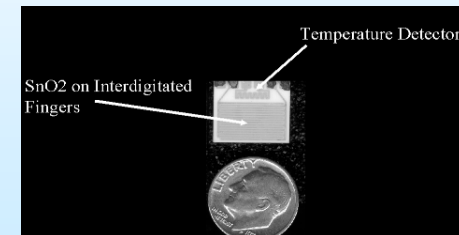


50 Hp Gas Turbine

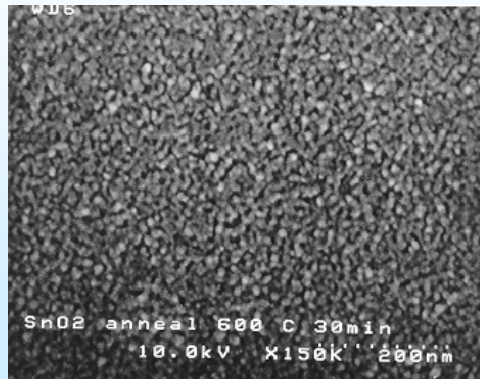


Industry Standard
Continuous Emission
Monitoring Equipment

**REPLACE INSTRUMENT RACK
SIZED SYSTEM WITH DIME SIZE
SENSOR AND ACCOMPANYING
ELECTRONICS**



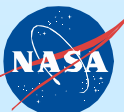
**Nanocrystalline
Tin Oxide**



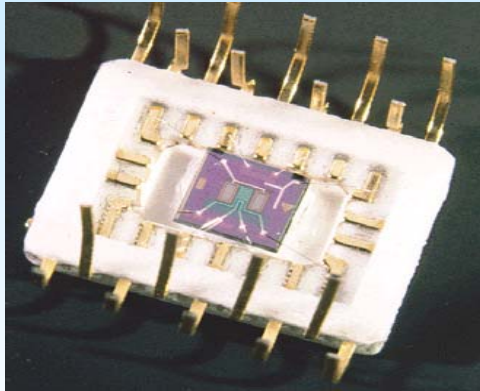
SENSOR SYSTEM DEVELOPMENT

- EACH SENSOR PLATFORM PROVIDES QUALITATIVELY VERY DIFFERENT TYPES OF INFORMATION ON THE ENVIRONMENT
- SENSOR ARRAY VARIES WITH APPLICATION/MICROFABRICATION TECHNIQUES MANDATORY
- BASIS CHEMICAL SENSOR FEATURES:
 - RESPONSE TIME, SENSITIVITY, SELECTIVITY, STABILITY
 - BATCH FABRICATION, PROCESSING REPRODUCIBILITY, CONTROL OF STRUCTURE
 - TAILOR SENSOR SYSTEM FOR THE APPLICATION
- SUPPORTING TECHNOLOGIES NECESSARY
 - PACKAGING (OFTEN UP TO 70% OF OVERALL SENSOR COST)
 - SIGNAL CONDITIONING AND PROCESSING
 - SOFTWARE (E.G. NEURAL NET PROCESSING, MODELING)
 - POWER AND COMMUNICATION

See for example: G. W. Hunt,
C.C. Liu, D. Makel,
Microfabricated Chemical
Sensors For Aerospace
Applications,
MEMS Handbook Second ed
Design and Fabrication, CRC
2006, Press LLC, Boca Raton
Ch. 11.



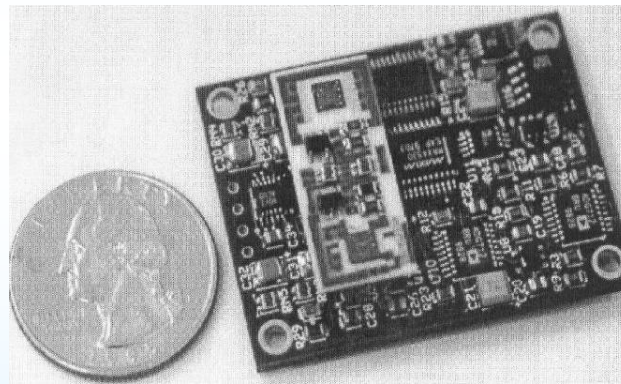
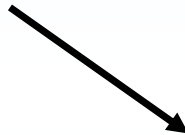
MINIATURIZED SMART LEAK SENSOR SYSTEM



**MICROFABRICATED HYDROGEN
SENSOR**



**HYDROGEN SENSORS ON SPACE
SHUTTLE**



PROTOTYPE HYDROGEN/OXYGEN SENSOR SYSTEM WITH ELECTRONICS

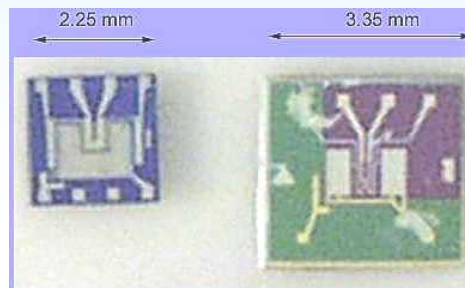


**DEMONSTRATE STAND-ALONE SMART LEAK DETECTION SYSTEM
WITH A SURFACE AREA THE SIZE OF POSTAGE STAMP**

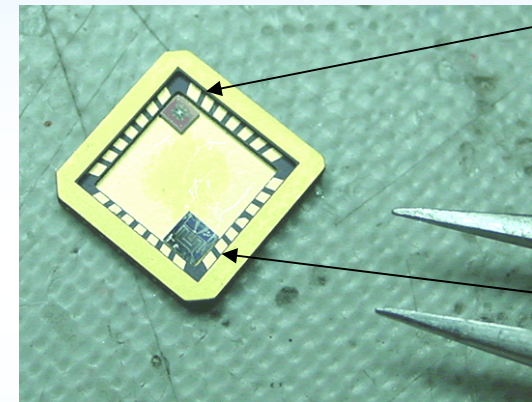
“LICK AND STICK” LEAK SENSOR SYSTEM

- COMBINE FUEL (HYDROGEN, HYDROCARBON) WITH OXYGEN IN AN ARRAY: DETERMINE EXPLOSIVE COMBINATIONS
- SELF-CONTAINED SYSTEM WHICH CAN BE IMPLEMENTED WHEREVER, WHENEVER NEEDED WITHOUT REWIRING OR SIGNIFICANT POWER DRAIN TO THE VEHICLE
- ON-GOING ACTIVITY: DECREASE SIZE AND POWER OF SENSORS/ELECTRONICS

**FIRST STEP:
COMBINE
SMALLER
SENSORS WITH
SMALLER,
SMART
ELECTRONICS**

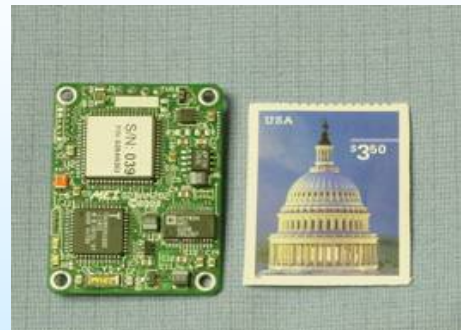


HYDROGEN SENSOR SIZE REDUCTION



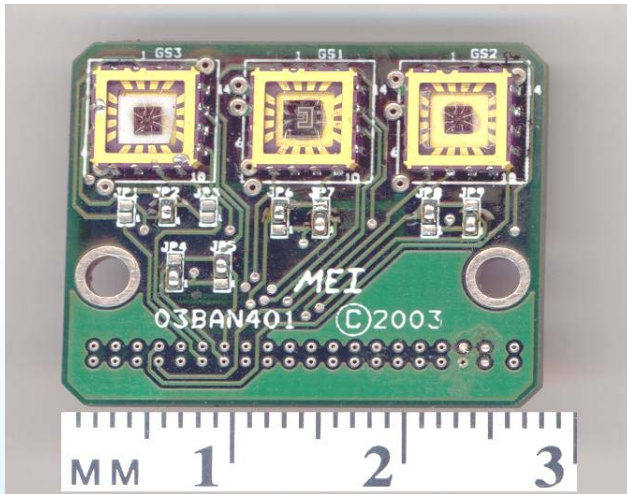
**HYDROGEN AND OXYGEN
SENSORS PACKAGED ON
SAME CHIP**

**MINIATURIZED
ELECTRONICS**

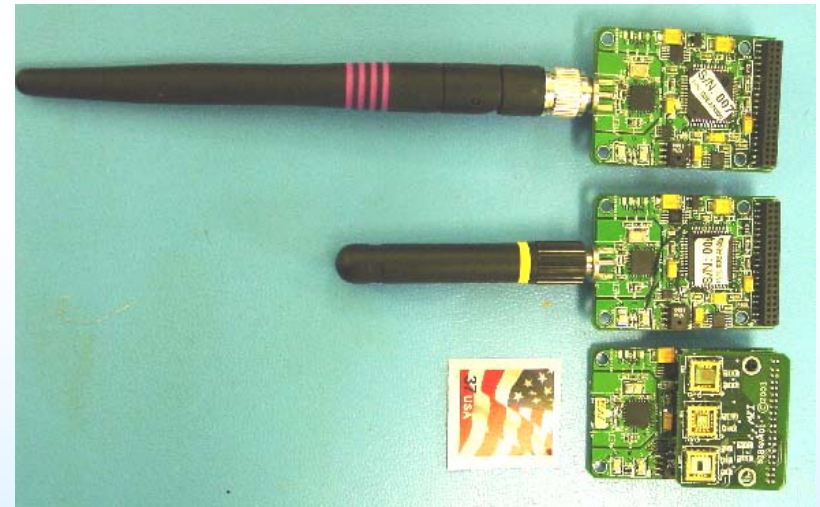


“LICK AND STICK” LEAK SENSOR SYSTEM

- **SENSORS, POWER, AND TELEMETRY SELF-CONTAINED IN THE NEAR THE SIZE OF A POSTAGE STAMP**
- **MICROPROCESSOR INCLUDED/SMART SENSOR SYSTEM**
- **VERIFY SYSTEM COMPATIBILITY WITH SPACE APPLICATIONS**
- **ADAPTABLE CORE SYSTEM WHICH CAN BE USED IN A RANGE OF APPLICATIONS**
- **MULTIPLE CONFIGURATIONS AVAILABLE**



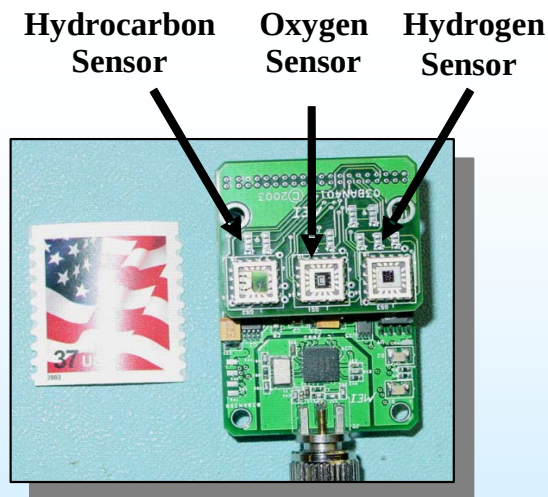
**“Lick and Stick” Leak
Detection Electronics and
Three Sensors**



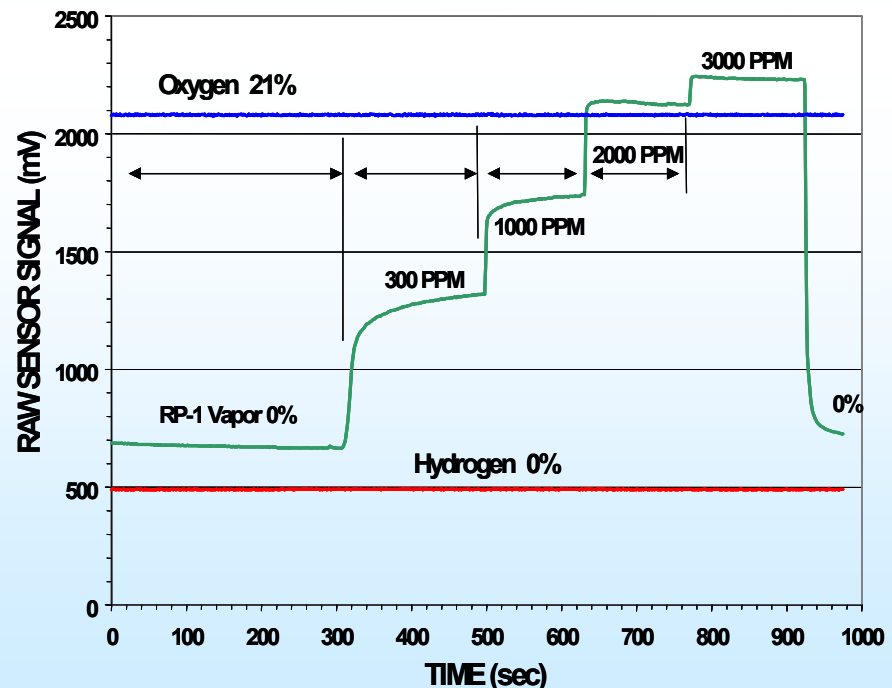
**System configured with
different wireless antennae.**

“LICK AND STICK” LEAK SENSOR SYSTEM DEMONSTRATION

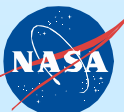
- WIRELESS DEMONSTRATION OF 3 SENSOR SYSTEM ACHIEVED
- BASELINE: ZIRCONIA BASED O₂ SENSOR (ALTHOUGH NAFION BASED ROOM TEMPERATURE SYSTEM BEING MATURED FOR USE)
- LONGEVITY OF SENSOR SYSTEM LIFE ON A BATTERY IS A LIMITATION IN SOME APPLICATIONS
- MOVE FROM HIGH TEMPERATURE SENSOR TECHNOLOGY TO LOWER TEMPERATURE SENSORS
- BEING QUALIFIED FOR CREW LAUNCH VEHICLE APPLICATIONS (HARDWIRED) FOR HYDROGEN DETECTION ONLY



MEI Makel Engineering Inc.



Glenn Research Center



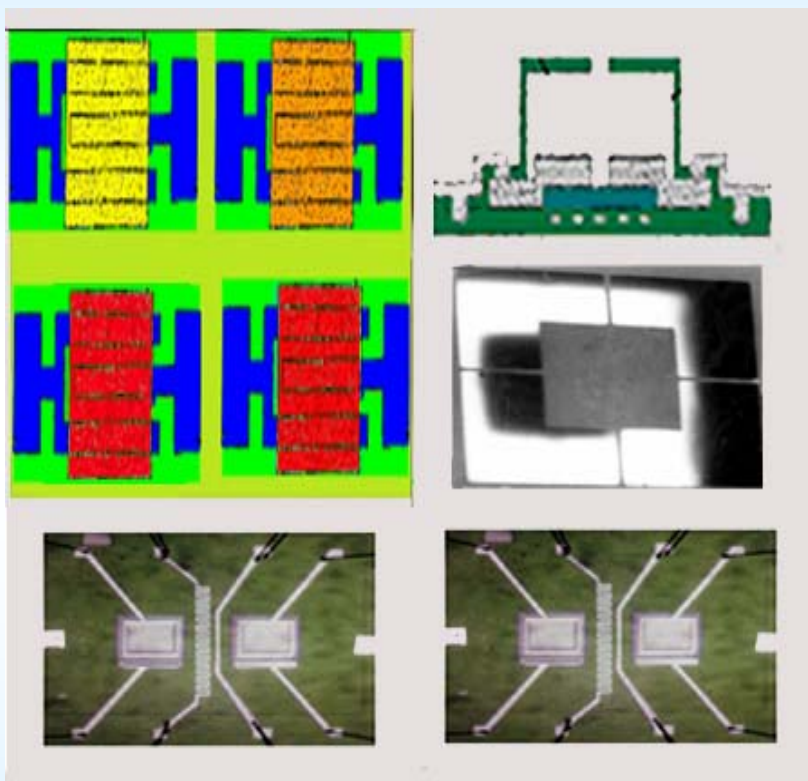
HIGH TEMPERATURE GAS SENSOR ARRAY HIGH TEMPERATURE ELECTRONIC NOSE

**SnO₂
Resistor**

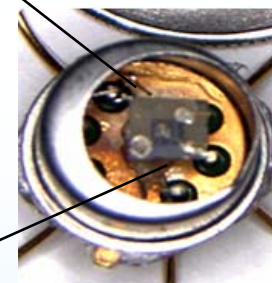
**TiO₂
Resistor**

**Electrochemical Oxygen
Sensor**

**Selectively
Filtered
SnO₂ Resistors**



**SiC-Based
Pressure Sensor**



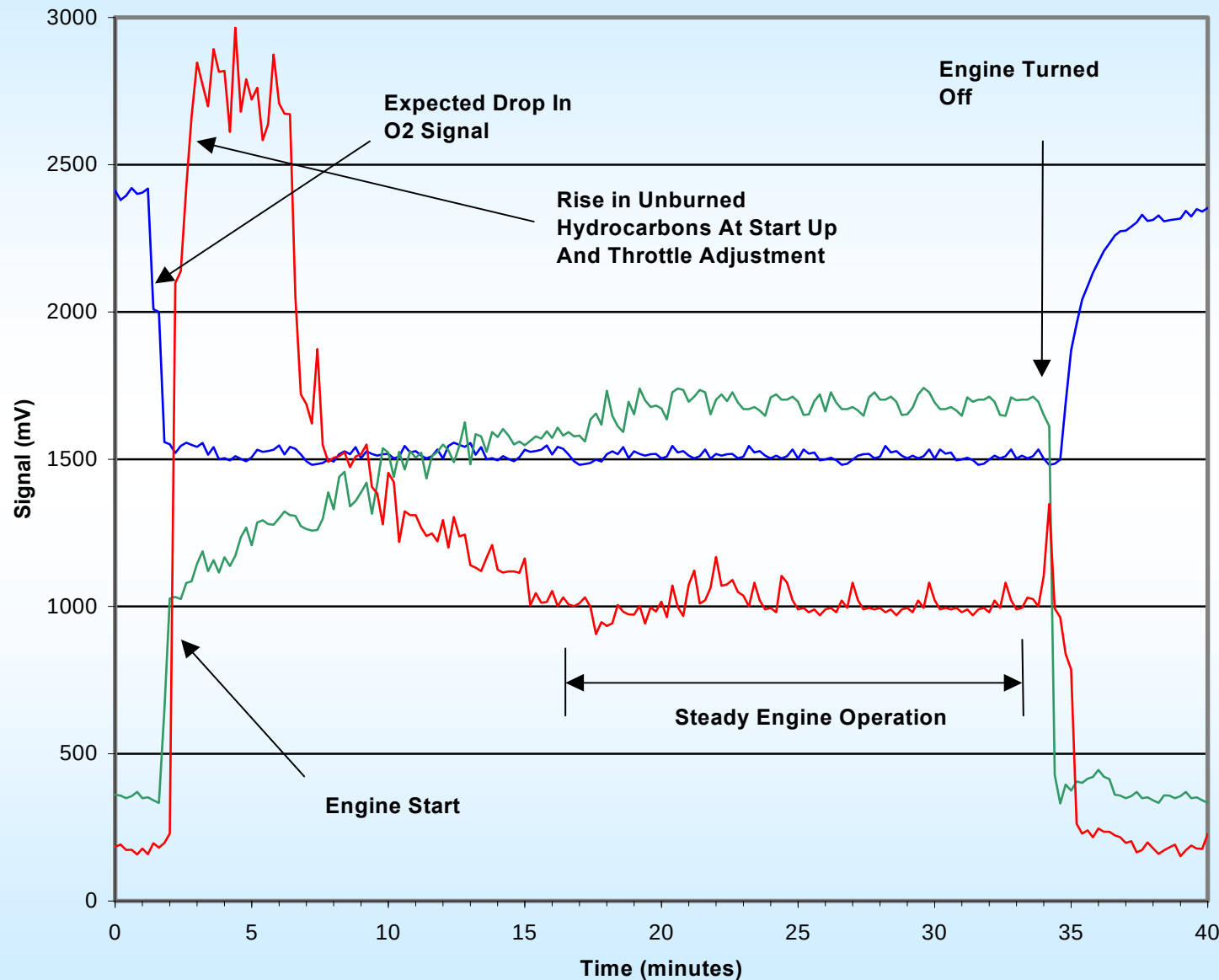
**Metal-SiC
Schottky diodes**

**Metal-Reactive Insulator
SiC Schottky diodes**

Harsh Environment Demonstration Testing



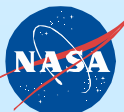
1.9 liter, four cylinder HCCI at U.C. Berkeley (propane/air)



Makel Engineering
Inc.

Exhaust Gas Temperature
Phi= 0.35
O2=14%
NOx<5 PPM
CO =1400 PPM
UHC =1200 PPM

— Oxygen Sensor
— SnO2 Sensor
— SiC Hydrocarbon Sensor

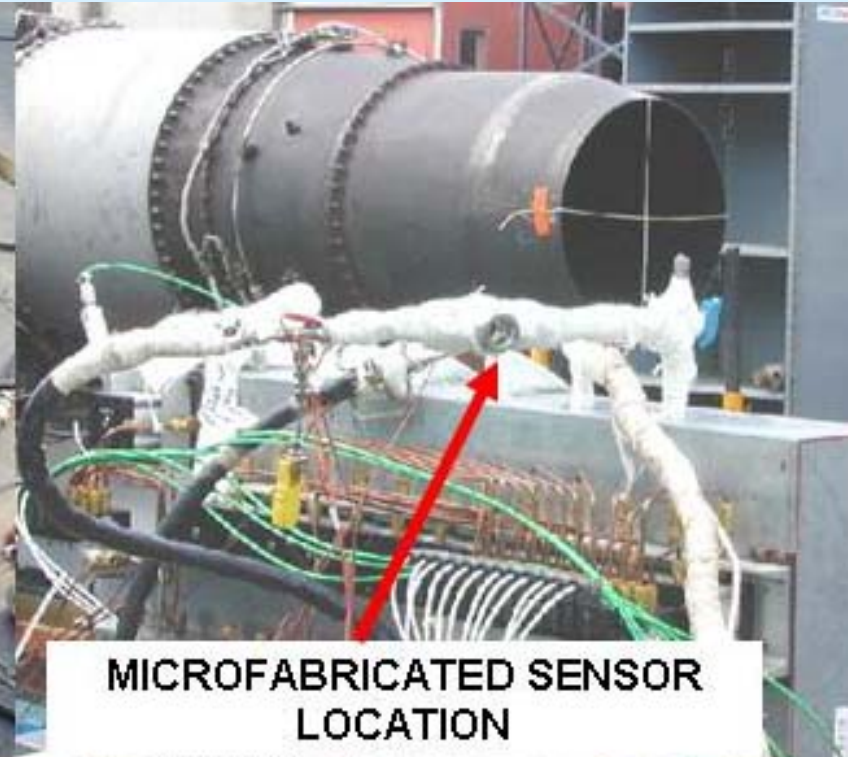


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MICRO SENSORS TESTED AT OUTLET OF THE JT-12 JET ENGINE



Rake Sampling System At The Outlet Of The JT-12 Jet Engine.



Location Of The Sensors In The Flow Stream Of The Rake

HIGH TEMPERATURE OPERATIONAL CAPABILITY AND OF THE SENSORS ALLOW PLACEMENT SIGNIFICANTLY CLOSER TO THE ENGINE OUTLET THAN TRADITIONAL EQUIPMENT.

Micro-Fabricated Gas Sensors for Low False Alarms

2005 R&D 100 AWARD WINNER

NASA 2005 TURNING GOALS INTO REALITY AA's CHOICE AWARD

FEATURES

•MICROFABRICATED CO/CO₂ GAS SENSOR ARRAY

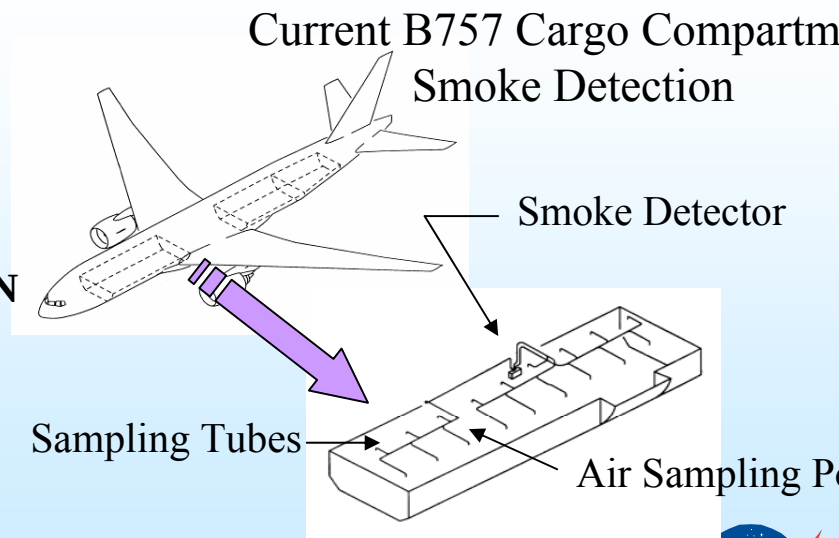
- AIM TO DECREASE FALSE ALARM RATE WHICH IS AS HIGH AS 200:1
- CENTRAL TO APPROACH
- NANOCRYSTALLINE MATERIALS (IN CO SENSOR) PRODUCE MORE SENSITIVE, STABLE SENSORS
- TWO APPROACHES TO CO₂ DETECTION
- MINIMAL SIZE/WEIGHT/POWER

•CHEMICAL GAS SENSORS PROVIDE GASEOUS PRODUCT-OF-COMBUSTION INFORMATION

- SENSOR ARRAY CAN DETECT RANGE OF GAS SPECIES
- TO BE COMBINED WITH INTELLIGENT SOFTWARE FOR PATTERN RECOGNITION

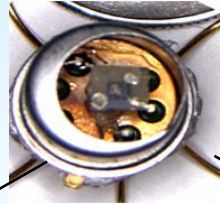
•BENEFITS

- DISCRIMINATE FIRES FROM NON-FIRES

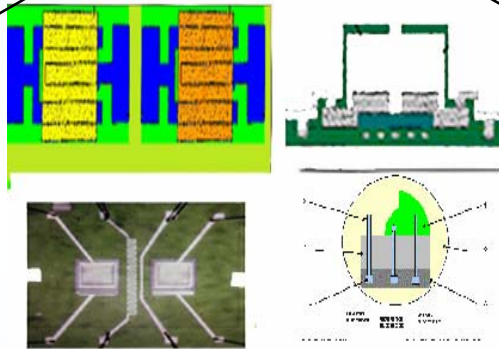


OVERALL FIRE DETECTION APPROACH FOR SPACE APPLICATION

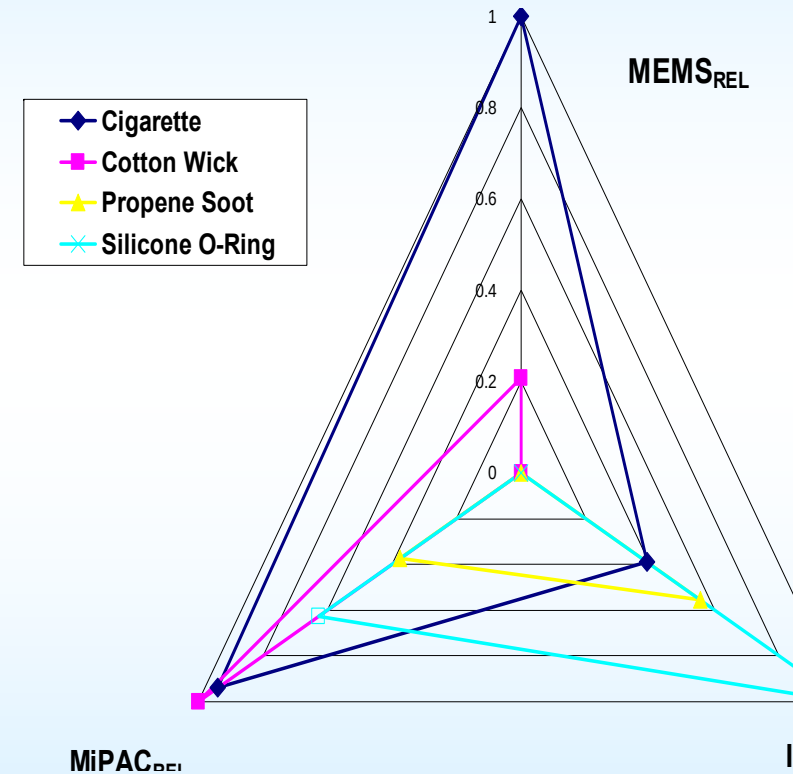
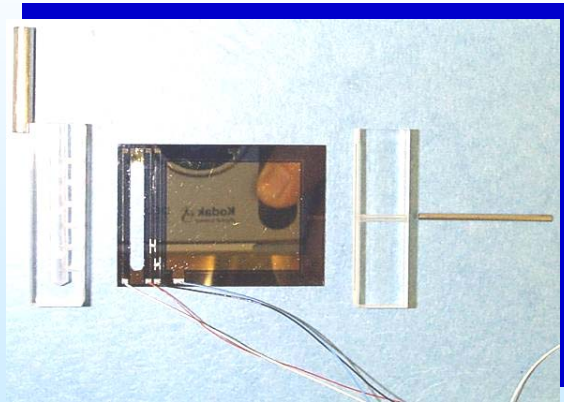
- COMBINED MEMS-BASED CHEMICAL SPECIES AND PARTICULATE
- ORTHOGONAL DETECTION AND CROSS-CORRELATION SIGNIFICANTLY REDUCES FALSE ALARMS



**MEMS-Based
Chemical
Species
Detection**

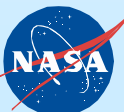


**MEMS-Based
Particulate
Detector**

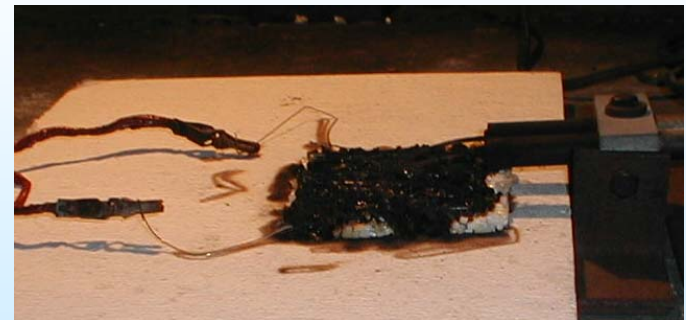


Makel Engineering, Inc.

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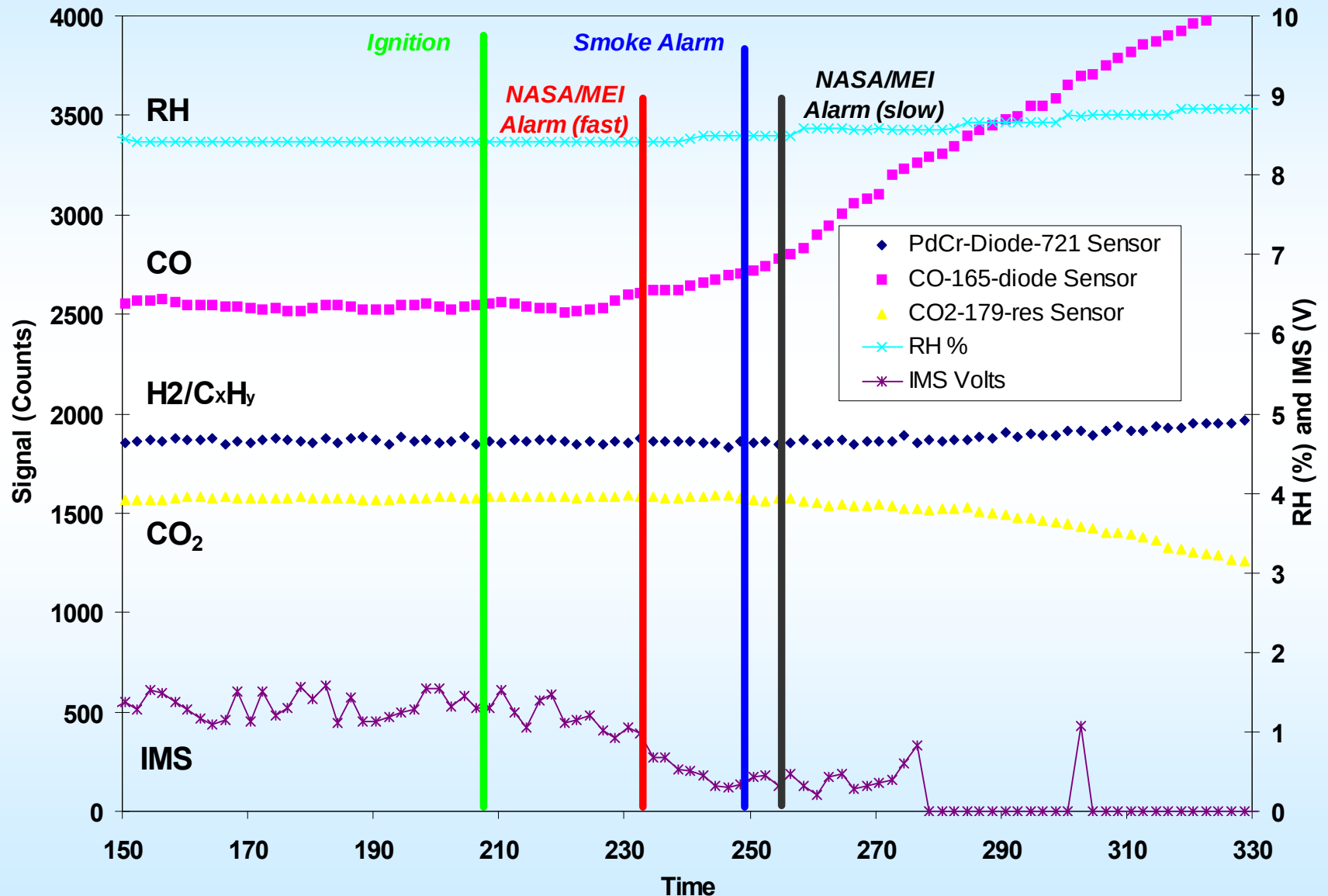
FAA Cargo Bay Fire Simulation Testing Boeing 707 luggage compartment and the FAA “Biscuit”



FAA Cargo Bay Fire Testing

No False Alarms/Consistent Detection of Fires

Transitioning to Space Fire Applications



SUPPORTING TECHNOLOGIES

**SUPPORTING TECHNOLOGIES OFTEN DETERMINE
SUCCESS OF SYSTEM**

HARSH ENVIRONMENT SYSTEMS

High Temperature Wireless Development

OBJECTIVES:

- HIGH TEMPERATURE WIRELESS TELEMETRY, DISTRIBUTED ELECTRONICS OVER A BROAD OPERATING RANGE

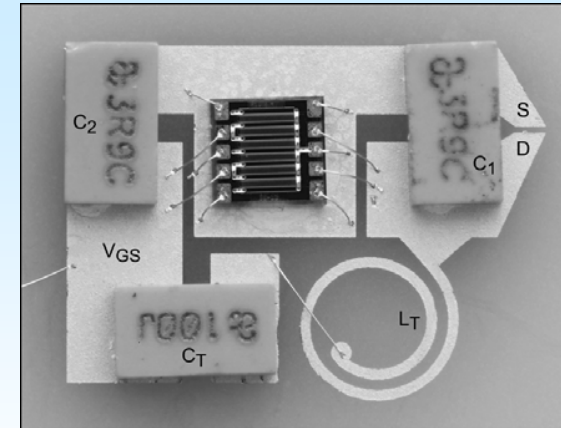
TECHNICAL CHALLENGES:

- DEVELOPMENT OF RELIABLE HIGH TEMPERATURE TELEMETRY ELECTRONICS, POWER SOURCES, REMOTE COMMUNICATION ELECTRONICS, AND PACKAGING

GOALS SUPPORTED:

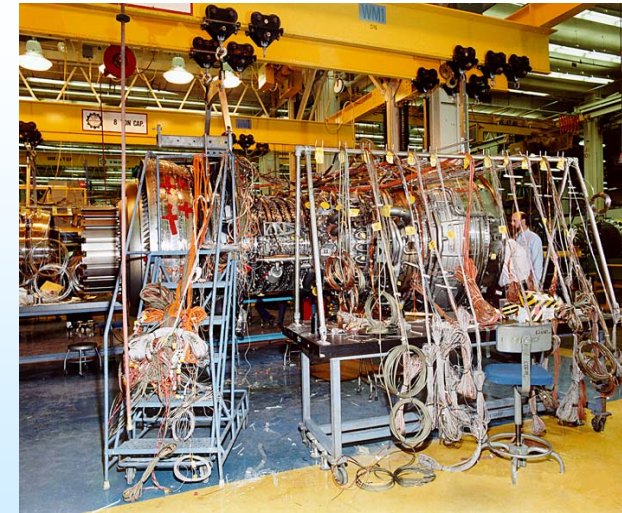
- ENHANCE PERFORMANCE
- SIGNIFICANTLY REDUCE COST

PROVIDE DATA TRANSFER IN HARSH ENVIRONMENTS IMPROVING RELIABILITY AND ENABLING NEW CAPABILITIES



Prototype Oscillator Circuit

Example: Gas Turbine Engine Development
Requires Extensive Instrumentation
Yielding Extensive Wiring Complexity

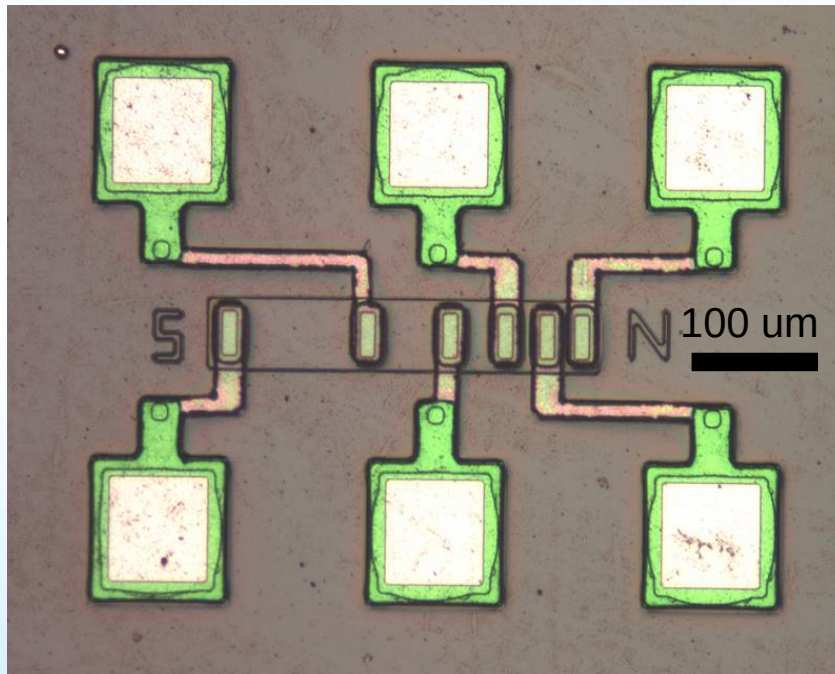


Wires from 1000 Sensors

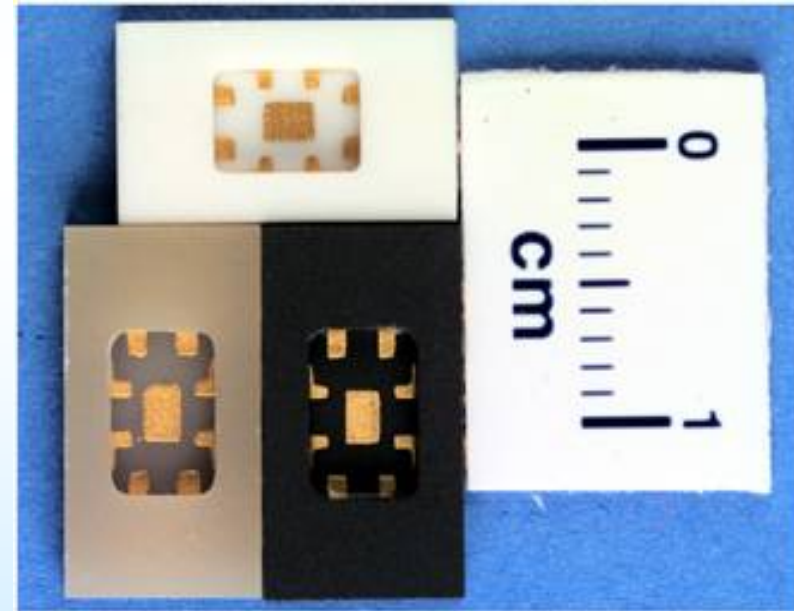
Previous Key NASA Glenn Advancements

Key fundamental high temperature electronic materials and processing challenges have been faced and overcome by systematic basic materials processing research (fabrication and characterization).

500 ° C Durable Metal-SiC Contacts
(R. Okojie, 2000 GRC R&T Report)



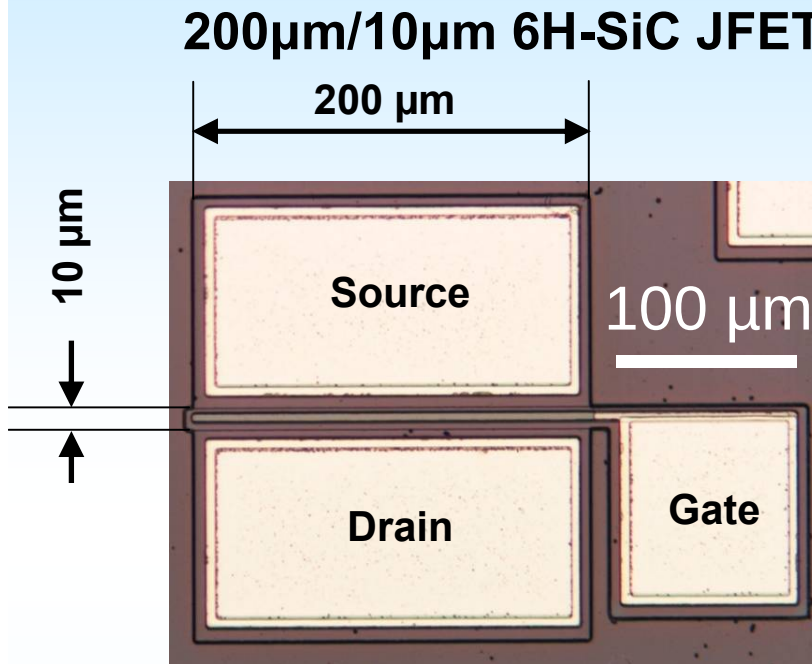
500 ° C Durable Chip Packaging
And Circuit Boards
(L. Chen, 2002 GRC R&T Report)



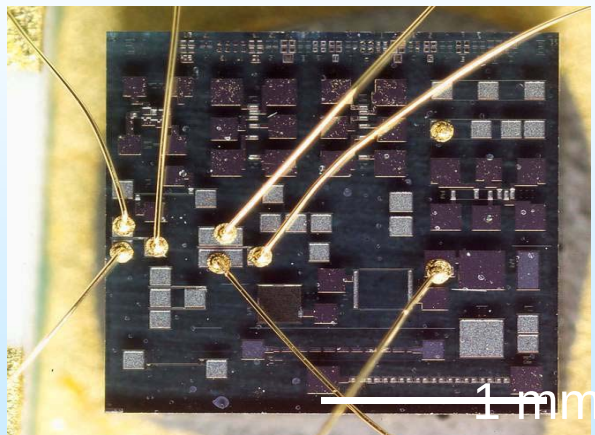
Additional advancements in device design, insulator processing, etc. also made.

6H-SiC Junction Field Effect Transistor (JFET) Fabricated by NASA Glenn Research Center

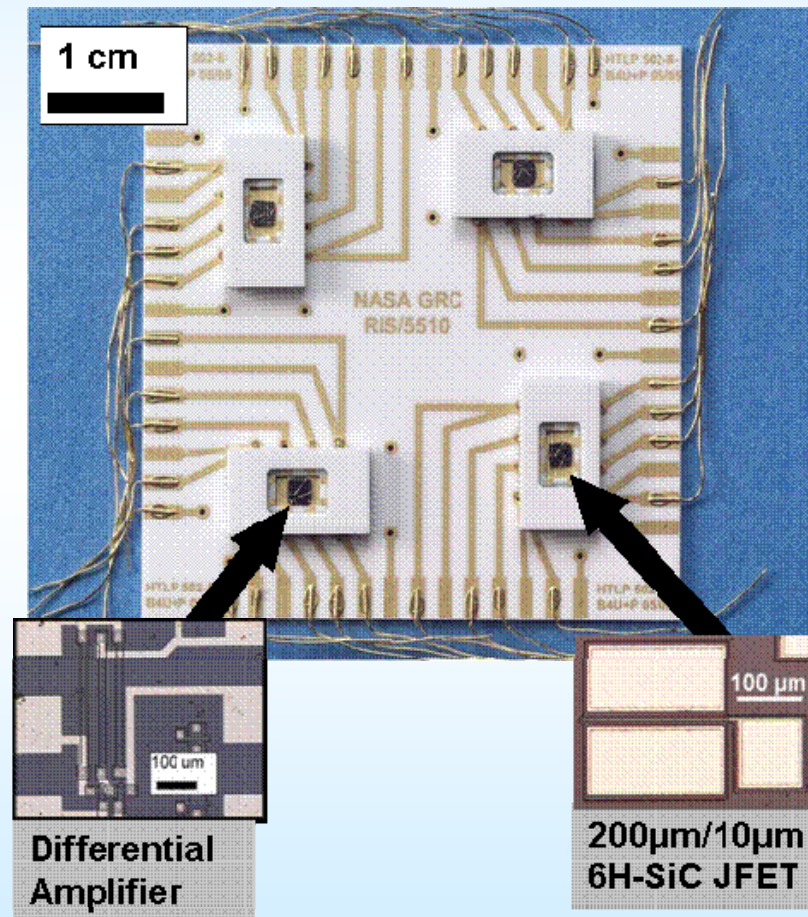
Tech Accomplishment
(IVHM v1.5 MS 1.3)



Packaged with bond wires



Multiple devices in Ceramic Packaging



Glenn Research Center



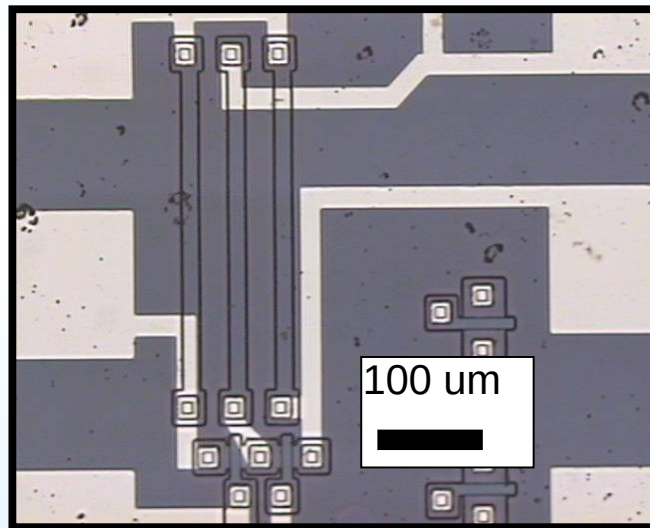
NASA Glenn Silicon Carbide Differential Amplifier

Tech Accomplishment
(IVHM v1.5 MS 1.3)

World's First Semiconductor IC to Surpass
4000 Hours of Electrical Operation at 500 ° C

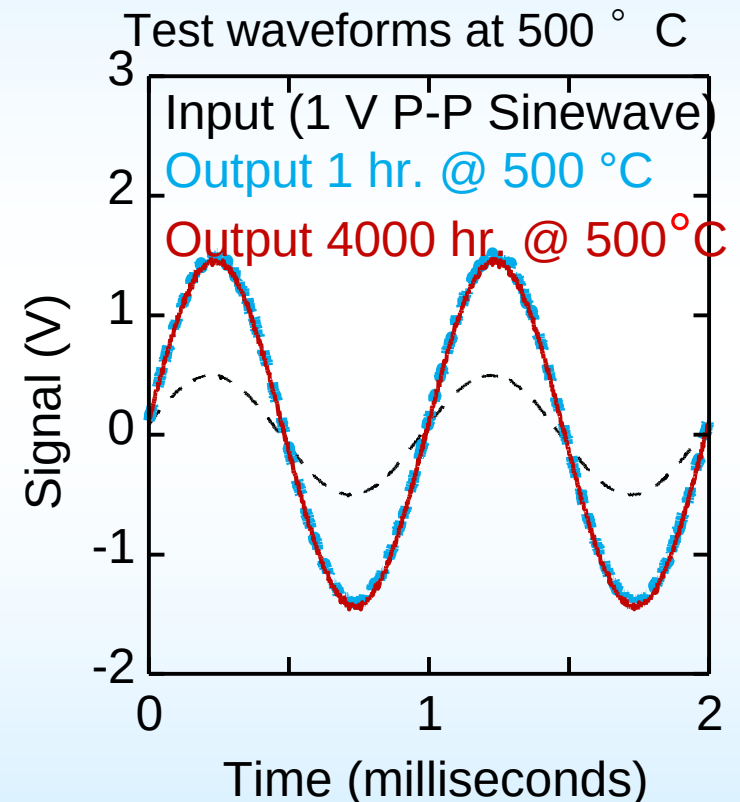
Demonstrates CRITICAL ability to interconnect transistors and other components (resistors) in a small area on a single SiC chip to form useful integrated circuits that are durable at 500 ° C.

Optical micrograph of demonstration
amplifier circuit before packaging



2 transistors and 3 resistors
integrated into less than half a
square millimeter.

Single-metal level interconnect.

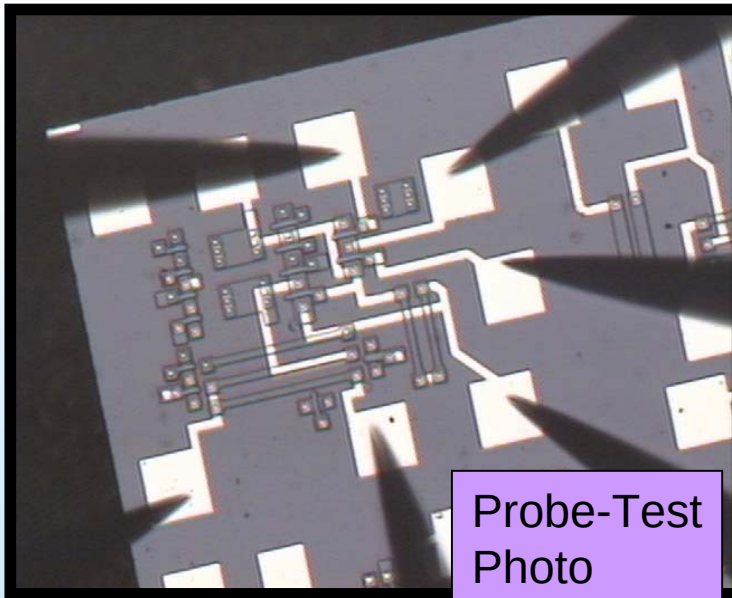
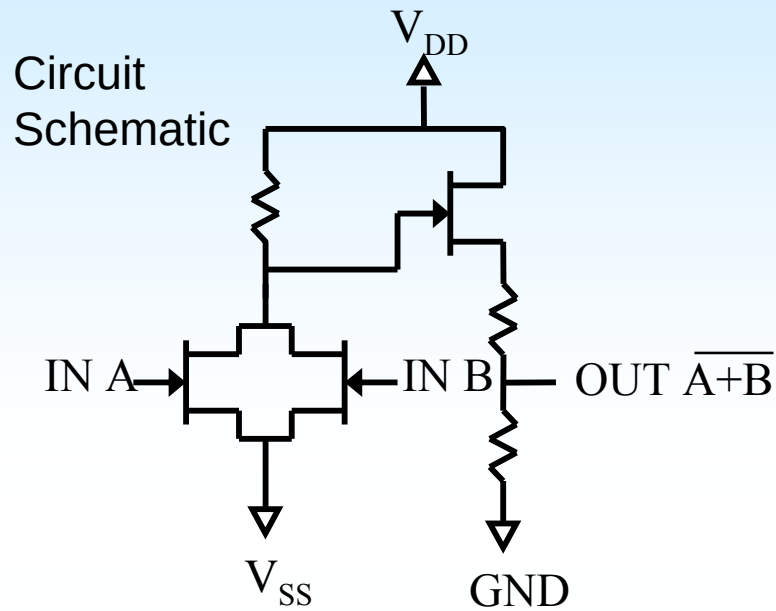


Less than 5% change in
operating characteristics during
4000 hours of 500 ° C operation.

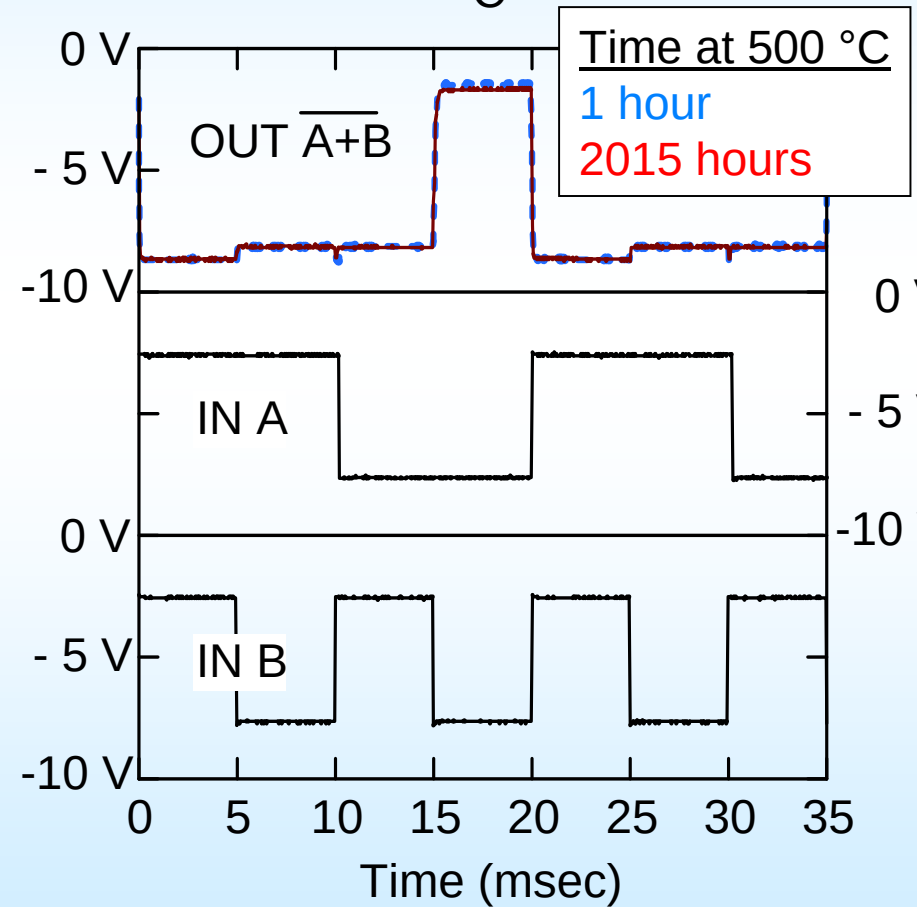
NASA Glenn SiC JFET NOR Gate IC

Tech Accomplishment
(IVHM v1.5 MS 1.3)

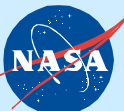
**World's First Semiconductor Digital IC to
Surpass 2000 hours of 500° C Operation**



Waveforms of **packaged**
NOR (= "Not OR") gate at 500
° C



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***SIGNIFICANCE OF RECENT ELECTRONICS RESULTS
THE BASIC HARDWARE TOOLS FOR HIGH TEMPERATURE DATA PROCESSING
HAVE BEEN FABRICATED***

- ◆ **THESE RESULTS HAVE BEEN THE SUBJECT OF A HIGH LEVEL OF VISIBILITY
E.G. NASA TOP 10 DISCOVERY STORIES FOR 2007**
- ◆ **DURABLE HIGH TEMPERATURE IC'S WILL ENABLE IMPORTANT NEW
CAPABILITY**
 - Enabled by fundamental electronic materials research.
 - World record IC durability at 500 ° C (> 400-fold improvement).
 - Inherently up-scalable to high circuit complexity while remaining physically small.
- ◆ **THIS DEMONSTRATION SHOWS THAT IT IS NOW POSSIBLE TO CONSTRUCT
MORE COMPLEX CIRCUITS OPERATING AT 500 °C AND MINIATURIZED.**
- ◆ **LOGIC GATES GENERATE FLIP-FLOPS THAT CAN GENERATE STATE-
MACHINES TO ENABLE:**
 - Creation Of Control Electronics For An “Intelligent” Fixed Or Mobile Agent
 - The Configuration Of Intelligent Data Transmission Methods Allowing For Unambiguous Demodulation Of Signals Uniquely Associated With Each Sensor/Transmitter In A Network.

**OBJECTIVE : TO MOVE TOWARD HIGHER DEGREES OF COMPLEXITY
ALLOWING WIRELESS TRANSMISSION AND HARSH ENVIRONMENT
SMART SENSOR SYSTEMS**

Overall Approach:

Smart Systems in High Temperature Environments

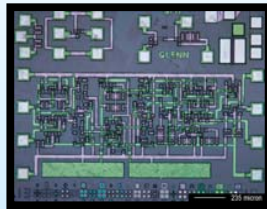
**Milestone: Demonstrate High Temperature Sensing, Wireless Communication,
and Power Scavenging for Propulsion Health Management
8/30/2011**

**Metric: Demonstrate integrated self powered wireless sensor system at 500
C with data transmission with operational life of at least 1 hr**

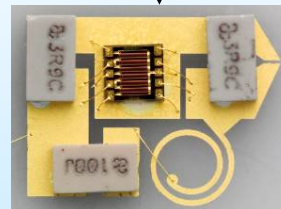
Significant wiring
exists with present
sensor systems



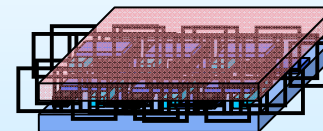
Allow Sensor Implementation by Eliminating Wires



World Record High
Temperature Electronics
Device Operation



High Temperature RF
Components



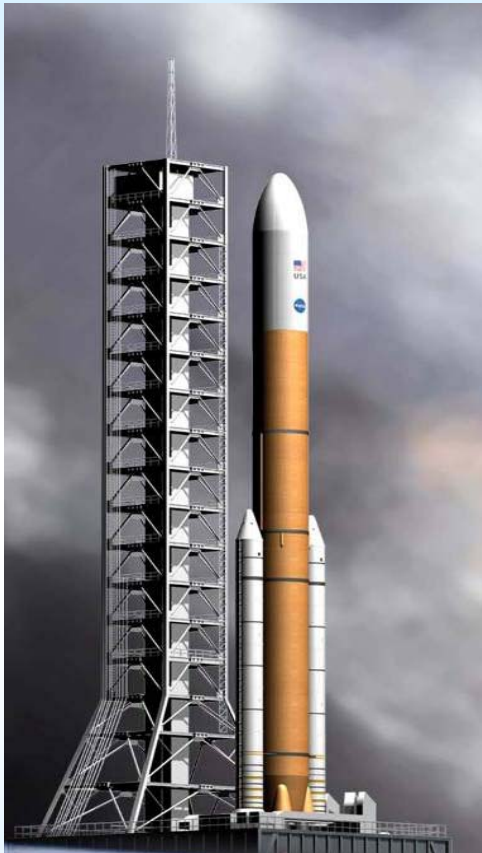
Energy Harvesting
Thin Film
Thermoelectrics

SENSOR SYTEM APPLICATION

SYSTEM TESTING TO MOBILITY

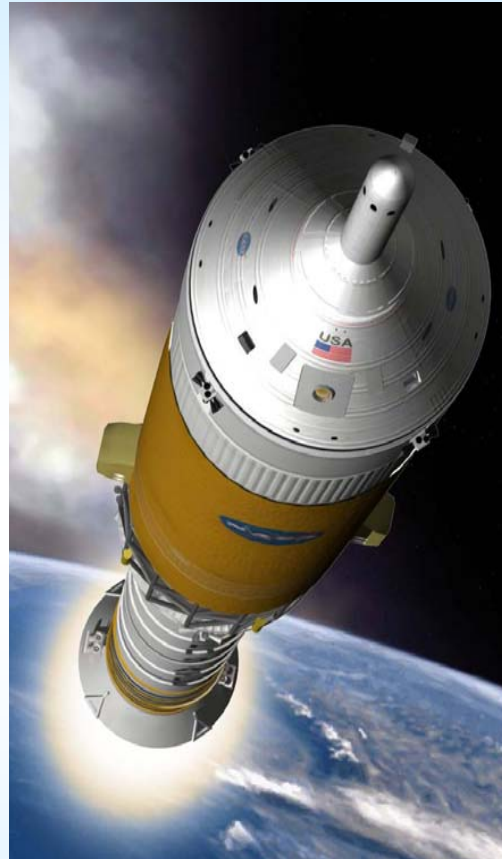
**FUTURE APPLICATIONS
LICK AND STICK SYSTEMS
BRIEF LIST OF LAUNCH, IN-SPACE, AND LUNAR APPLICATIONS**

LAUNCH



**Propellant Leaks
Toxic Gas Leaks**

IN-SPACE



**Propellant Fuel Leaks
Toxic Gas Leaks
Environmental Monitoring
Fire Detection
EVA**

LUNAR



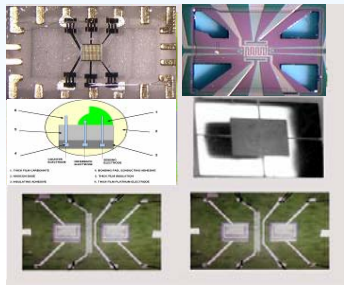
**Propellant Fuel Leaks
Toxic Gas Leaks
Environmental Monitoring
Fire Detection
EVA/ISRU Applications**

SUMMARY

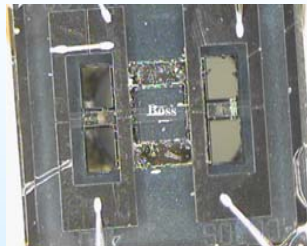
TOOLS TO ENABLE NEW MISSIONS

EXAMPLE POSSIBLE MISSION: Venus Integrated Weather Sensor (VIWS) System

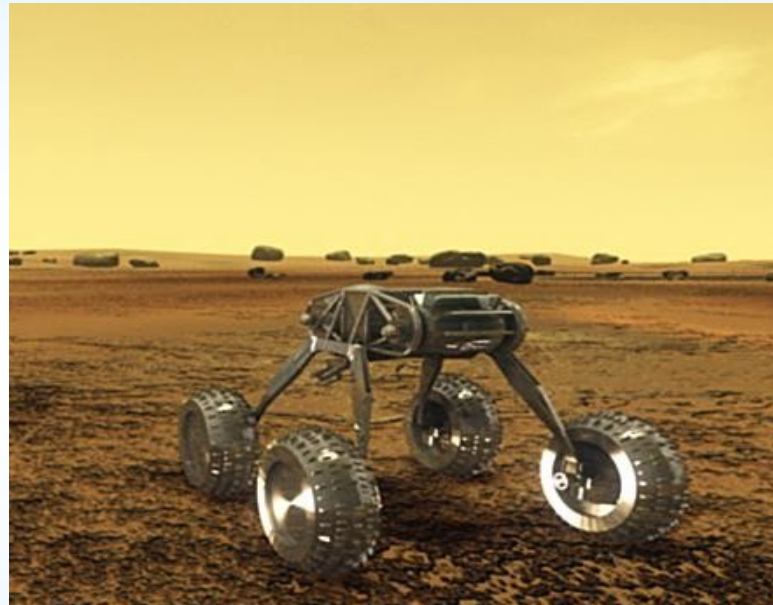
Sensor Suite to Monitor Venus Weather Conditions including: **Data Processing and Communication, Wind Flow, Seismic, Pressure/Temperature/Heat Flux, Chemical Environment**



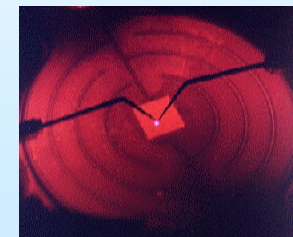
**HIGH TEMPERATURE
ELECTRONIC NOSE
(Chemical Species)**



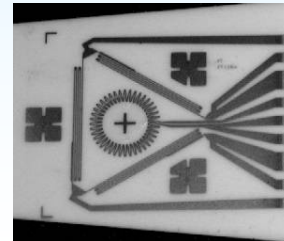
**Hi-g SiC
ACCELEROMETER
(Seismic Activities)**



**PRESSURE
SENSOR
(Pressure)**



**SiC ELECTRONICS
(Data Processing
and Com)**



**MULTIFUNCTIONAL
PHYSICAL SENSOR ARR
(Temperature, Heat Flux)**



**HOTPro
(Wind flow
Pressure
Temperature)**

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One Potential Vision: “Smart” Suit

- Development of a “Smart” Suit which has self-monitoring, caution and warning, and control capabilities with high levels of reliability, durability, and safety.
- Small, lightweight, low power sensor systems, with increased packaging flexibility, will improve the effectiveness and extensibility of the EVA suits.
- Seamless integration of sensors throughout EVA system improving reliability and capability without significantly increasing system wiring and power.
- Monitor Both Inside And Outside the EVA Suit for Astronaut Health and Safety\Suit Maintenance
 - **Inside: For Example, Monitor Suit CO2, O2, Flow to Allow Metabolic Measurements**
 - **Outside: For Example, Monitor Dust/Toxic Gas/Dangerous Conditions Before Brought Back Into Airlock Or Can Affect Astronaut Safety**
- Include Ability to Determine Astronaut Health by Monitoring of Breath

A “**Smart**” Suit
Needs To Monitor
Both Internal And
External Conditions



Breath Sensor System –
includes mouthpiece for
breath collection, Nafion
drying tube in sample
line, sensor manifold
with PDA interface, and
mini sampling pump

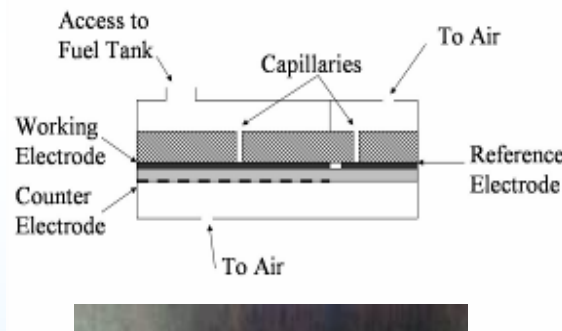


SENSOR SYSTEM IMPLEMENTATION

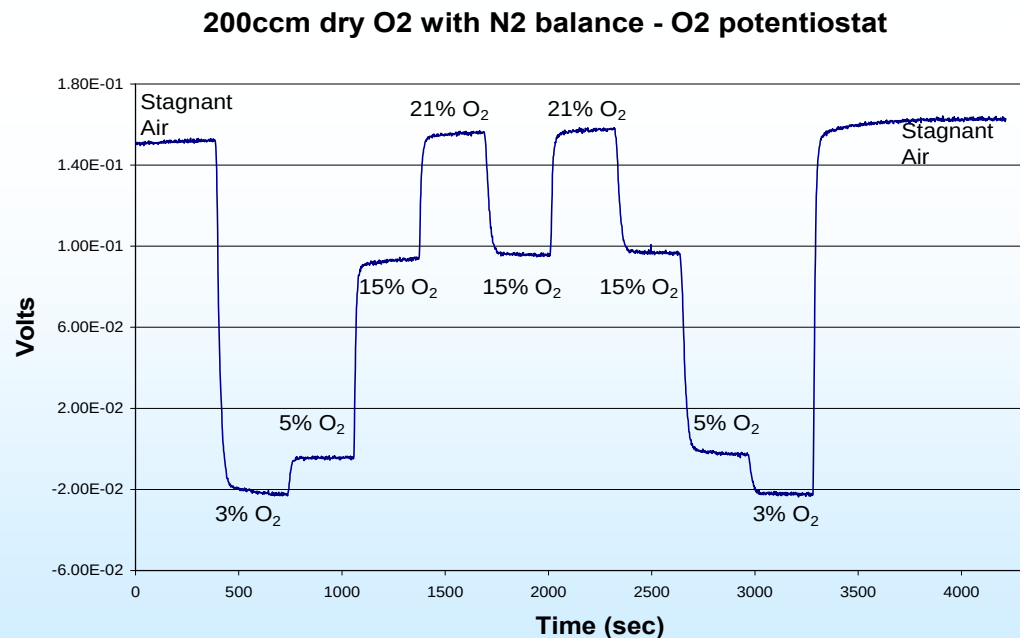
- **OBJECTIVE: A SELF-AWARE SYSTEM COMPOSED OF SMART COMPONENTS MADE POSSIBLE BY SMART SENSOR SYSTEMS**
- **SENSOR SYSTEMS ARE NECESSARY AND ARE NOT JUST GOING TO SHOW UP WHEN NEEDED/TECHNOLOGY BEST APPLIED WITH STRONG INTERACTION WITH USER**
- **SENSORS SYSTEM IMPLEMENTATION OFTEN PROBLEMATIC**
 - **LEGACY SYSTEMS**
 - **CUSTOMER ACCEPTANCE**
 - **LONG-TERM VS SHORT-TERM CONSIDERATIONS**
 - **SENSORS NEED TO BUY THEIR WAY INTO AN APPLICATION**
- **SENSOR DIRECTIONS INCLUDE:**
 - **INCREASE MINIATURIZATION/INTEGRATED INTELLIGENCE**
 - **MULTIFUNCTIONALITY/MULTIPARAMETER MEASUREMENTS/ORTHOGONALITY**
 - **INCREASED ADAPTABILITY**
 - **COMPLETE STAND-ALONE SYSTEMS (“LICK AND STICK” SYSTEMS)**
- **POSSIBLE LESSONS LEARNED**
 - **SENSOR SYSTEM NEEDS TO BE TAILORED FOR THE APPLICATION**
 - **MICROFABRICATION IS NOT JUST MAKING SOMETHING SMALLER**
 - **ONE SENSOR OR EVEN ONE TYPE OF SENSOR OFTEN WILL NOT SOLVE THE PROBLEM: THE NEED FOR SENSOR ARRAYS**
 - **SUPPORTING TECHNOLOGIES OFTEN DETERMINE SUCCESS OF A SYSTEM**

ROOM TEMPERATURE O₂ SENSOR TECHNOLOGY

- POSSIBLE REPLACEMENT FOR PRESENT LIQUID ELECTROCHEMICAL CELL TECHNOLOGY
- VIABILITY FOR SPACE BASED APPLICATIONS MUST BE VERIFIED THROUGH AN EXTENSIVE TEST PROGRAM.
- ONE SIGNIFICANT LESSON OF PREVIOUS ISS IMPLEMENTATION IS THAT TESTING RELEVANT ENVIRONMENTS OVER THE REQUIRED SENSOR LIFE IS MANDATORY
- FIND THE ISSUES ON THE GROUND OR LAB AND NOT AFTER IT HAS BEEN DEPLOYED ON THE VEHICLE IN SPACE
- TEST TO FAILURE AND ANALYZE THE FAILURE IF POSSIBLE



NAFION O₂ Sensor Structure



Preliminary NAFION O₂ Sensor Data

WHITE SANDS TEST FACILITY O2 SENSOR TESTING

- TESTING OCCURRED SIDE BY SIDE WITH EXISTING ISS SENSOR SYSTEMS FOR ISS ENVIRONMENTAL MONITORING AT WHITE SANDS TEST FACILITY
- TESTING OCCURRED OVER A RANGE OF PRESSURES AND O2 CONCENTRATIONS INTEGRATED WITH ELECTRONICS AND PRESSURE COMPENSATION
- REPEATED CYCLES OVER SEVERAL TEST PERIODS APPROXIMATED ~8 YEARS OF ISS OPERATION
- ACCURACY OF CALIBRATION, REPEATABILITY OF DATA, RESPONSE TIME WERE MAJOR OF EVALUATION CRITERIA
 - THIS IS A CRIT 1 (RELATED TO LIFE OF CREW) FUNCTION WITH STRICT CALIBRATION/PERFORMANCE REQUIREMENTS (+/-0.8%)
- MAJOR FINDING: SENSOR FAILURE MECHANISMS IDENTIFIED

Present commercial
ISS O2 sensors



NAFION Sensors in
manifold

NAFION Sensors
Control Electronics

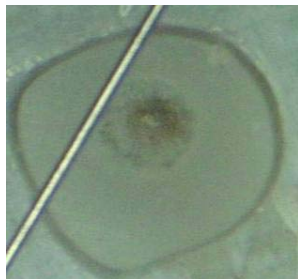
NAFION based oxygen sensor (left) and sensors during piggyback testing with NASA CSA-O2 system

PROGRAM RESULTS: ROOM TEMPERATURE O2 SENSORS CHANGES IN RESPONSE TO WHITE SANDS TESTING

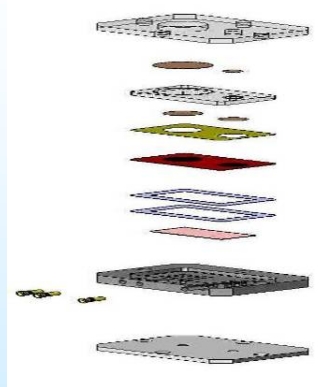
- 2.5 YEARS OF LIFE DURING BENCH TESTING/8 YEARS OF EQUIVALENT PRESSURE TESTING
- ROOM TEMPERATURE O2 SENSORS VIABLE FOR CRIT 1 ISS FUNCTION
 - SENSOR RESPONSE TIME, SENSITIVITY, CALIBRATION CAPABILITY
- LIFETIME OF O2 SENSORS WAS DETERMINED/PRIMARY GOAL OF PROJECT
 - FAILURE MODE WAS THAT THE O2 SENSORS STARTED DRIFTING OUTSIDE OF ALLOWABLE CRIT 1 TOLERANCES FOR O2 PERCENTAGE.



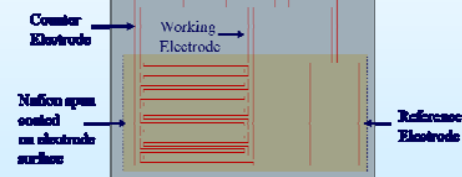
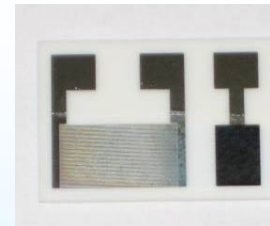
TESTING AT WHITE SANDS



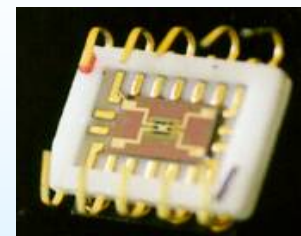
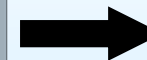
Example Failure Mechanism:
Diffusion Hole Breakdown



Present Sensor Structure



Sputtered Based Three Electrode System

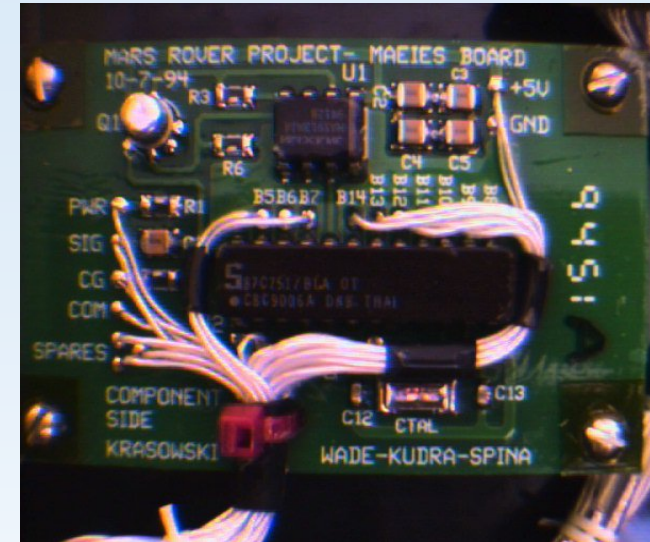


Batch Processed
MEMS based
Packaged Sensor

GRC Instrumentation For Space

Designed, Built, Programmed & Qualified In-House

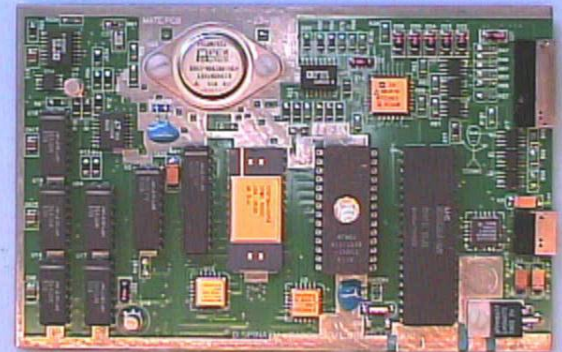
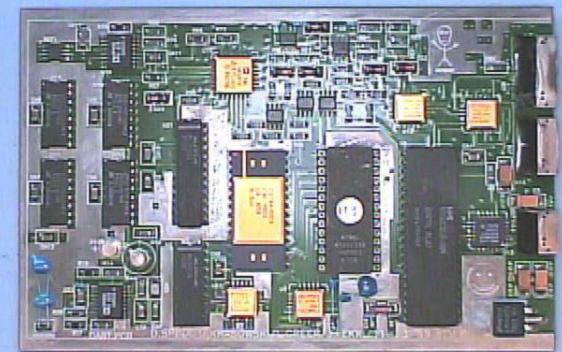
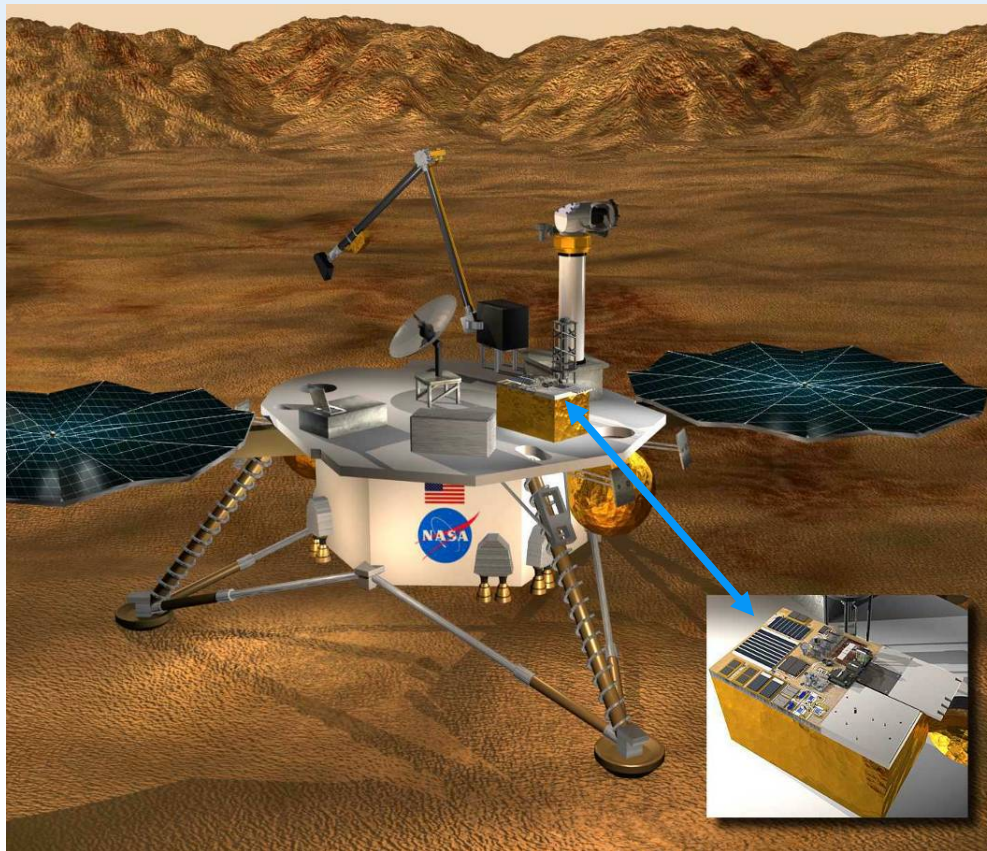
- 1996 – **Mars Pathfinder:**
Electronic hardware for Sojourner
 - Materials Adherence Experiment
Quartz Crystal Microbalance
 - Microprocessor based dust mass instrument with GRC specified, modified and characterized sensor
 - Electronics performed flawlessly determining that sensor saturated during unplanned secondary egress maneuver by JPL



GRC Instrumentation For Space

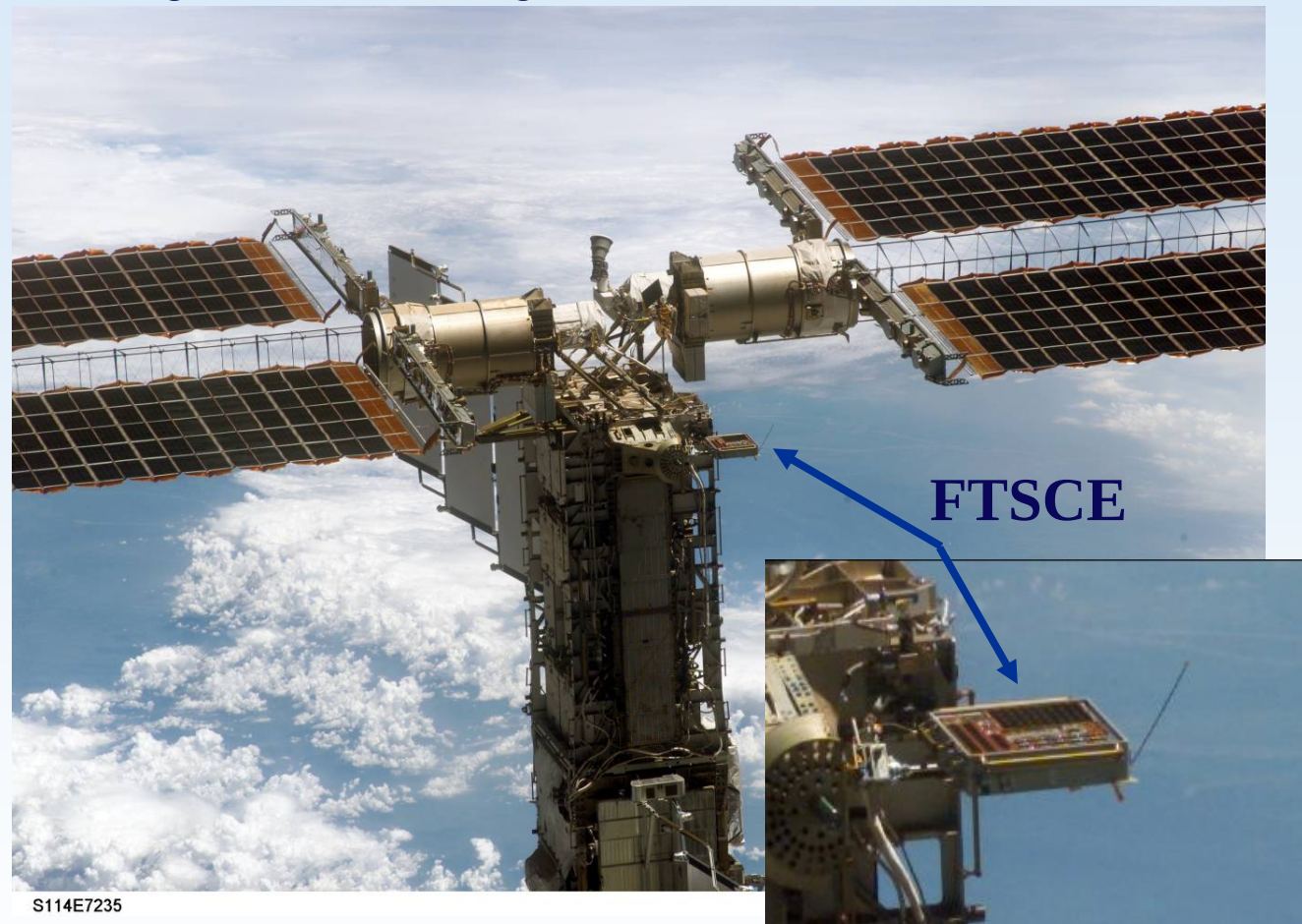
Designed, Built, Programmed & Qualified In-House

- 2001 – **Mars In-Situ Propellant Precursor (MIPP): Mars Array Technology Experiment and Dust Accumulation and Removal Technology (MATE & DART)**
 - Complete MIPP experiment top plate and circuit board set development to characterize multiple solar cell technologies and perform dust characterization and mitigation.



GRC Instrumentation For Space

Designed, Built, Programmed & Qualified In-House



Forward Technology Solar Cell Experiment (FTSCE) as part of 5th Materials on the International Space Station Experiment (**MISSE-5**)
Attached to the P6 Solar Panel Strut

GRC Instrumentation For Space

Designed, Built, Programmed & Qualified In-House

- July 2005 – **Forward Technology Solar Cell Experiment (FTSCE)** as part of 5th Materials on the International Space Station Experiment (**MISSE-5**)
 - **First Ever Active Experiment in MISSE Series**
 - 36 present and new solar cell technologies being exercised while directly exposed to Low Earth Orbit
 - Various temperature, sun position and radiometry sensors
 - **Operating through ground command radio link and autonomously through internal schedule, sun position and temperature**
 - **After a successful one year mission, FTSCE was returned to Earth and will be refurbished at GRC and flown again on MISSE-7**



GRC Mobile Sensor and Instrumentation Platforms

First Generation sensor platform:

- video
- packetized one way command link

Second Generation size shrink

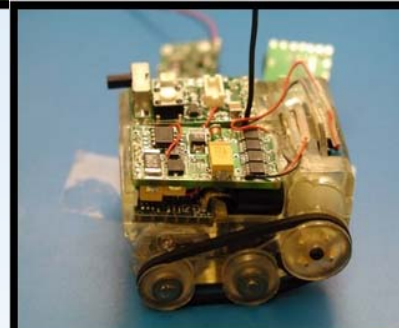
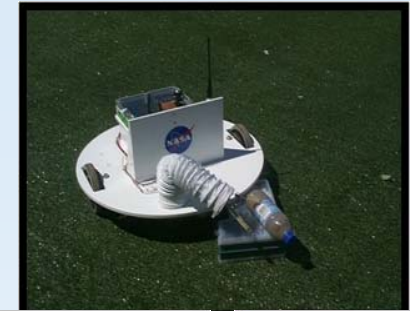
- with two way command and data path
- real time video
- multi-agent slotted network protocol

Telerobotics

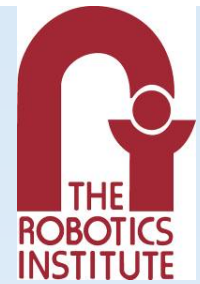
- Sensor Platform Area Network (SPAN)
- two agent control over the Internet
- streaming video
- each agent owns state vector of the other

Group Awareness and Behavior

- sensing the presence and position of other robot(s)
- near field sensing and communications channel
- mimics weakly electric fish



Highlander Lunar Rover Initiative



**“Cratos” - Lunar Rover Test Mule
Operating at NASA GRC**

**Carnegie Mellon Highlander Design
Benefits From Instrumentation and Motion
Control Electronics and Algorithms
Designed, Built and Tested in Cratos at
GRC.**

**NASA GRC and The Robotics Institute
at Carnegie Mellon University Co-
Developing Tracked Rovers Capable of
Autonomous Descent and Ascent of
Crater Walls**



CMU Highlander Rover

GRC and Case Western Reserve University Collaborative Effort In Biologically Inspired Robots: Whegs™

Whegs™ robots combine the advantages of combining wheels and legs and mimic the tripod gate used by the common Cockroach. Whegs allow a robot to climb over objects that are much higher than could be overcome using wheels.



An experimental Whegs in the laboratory

<http://biorobots.cwru.edu/projects/whegs/>



A step in environmentally hardening a Wheg

Whegs™ can also be hardened to swim on or under water and would make superb lake bottom crawlers

GRC and CWRU are hardening Whegs™ as potential Lunar, Mars and terrestrial sensor platforms

HOW DOES ONE MAKES SENSE OF ALL THIS SENSOR DATA QUALIFICATION

Sensor Implementation Considerations

For Ares I Crew Launch Vehicle

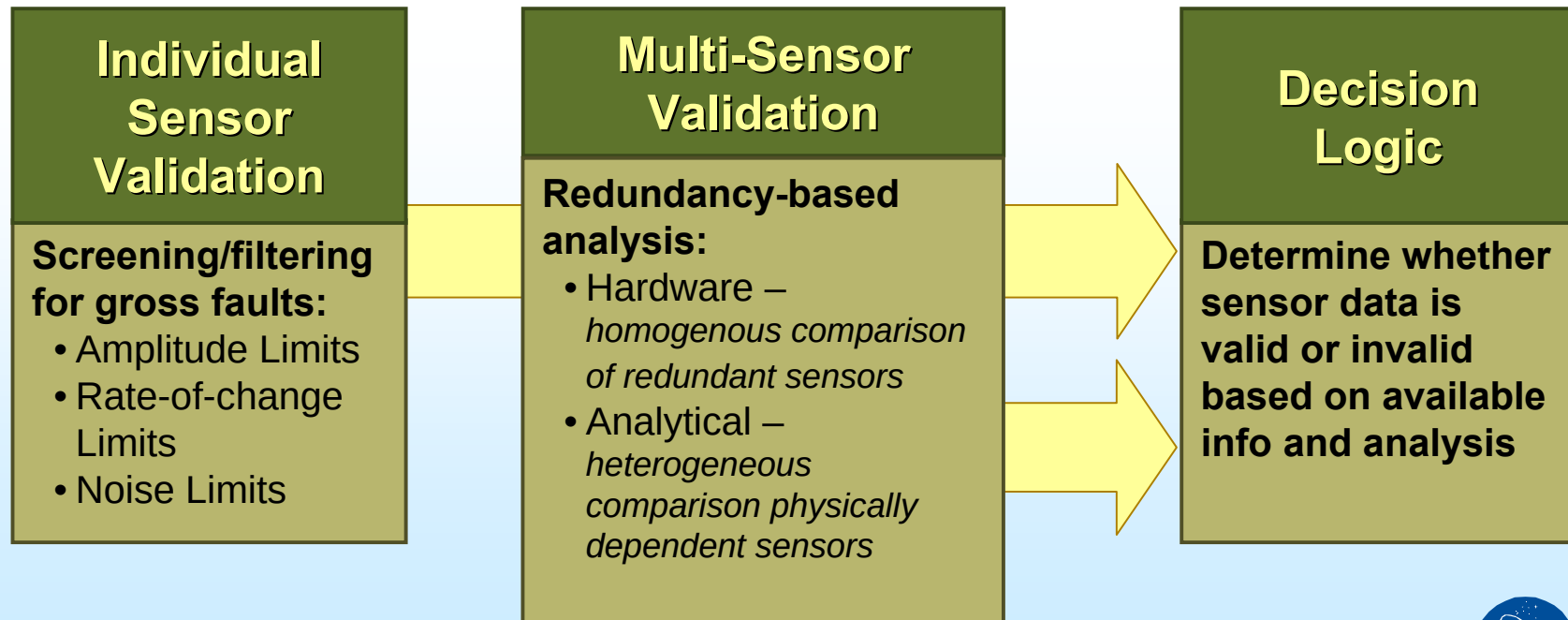
- **Each abort algorithm requires detection and confirmation of the condition**
 - Requires multiple hardware redundant measurements of the same property (homogeneous sensors)
- OR
- Require analytical redundancy with correlation of sensors measuring different, but related properties (heterogeneous sensors)
- **Flight Computer Software must provide data qualification on all flight critical sensors.**
 - Due to design constraints (e.g. cost, weight and heritage hardware) utilization of existing homogeneous and heterogeneous redundant sensors may be required

Data Qualification

- ◆ **Provides control and diagnostic systems with qualified sensor information**
 - Reduces probability that actions or operational decisions will be based upon faulty information
 - Increases confidence in measurements from unfailed sensors
- ◆ **Supports Launch Commit Criteria flexibility by identifying sensor faults versus system failures**
- ◆ **Screen faulty sensor data in performance data on-board prior to compression and transmission to ensure accurate information**
- ◆ **Software solution – requires no additional mass/weight**
 - Potential benefit of additional redundancy (through analytical relationships) without weight/complexity of additional sensors.
 - Adding more sensors may not be an option.

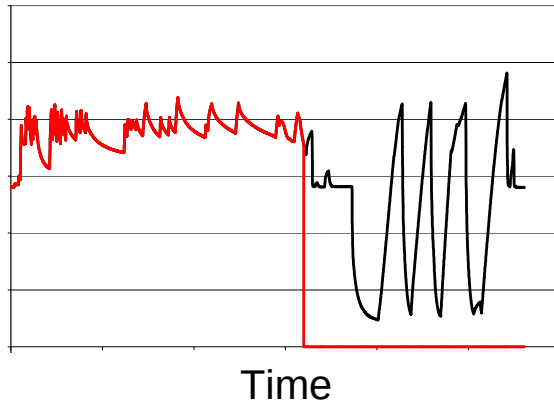
Solution $\Rightarrow$ Sensor Data Qualification System (SDQS)

- ◆ An algorithmic approach for continuously monitoring/analyzing sensor data at the flight computer to determine whether or not data values are within normal operating limits.
 - Sensor data is considered qualified if the SDQS determines that values are within the bounds of normal operation for a given sensor
 - Sensor data fails qualification if the SDQS determines that a values are outside the bounds of normal operation for a given sensor over a set time period
- ◆ Data qualification performed in three stages:

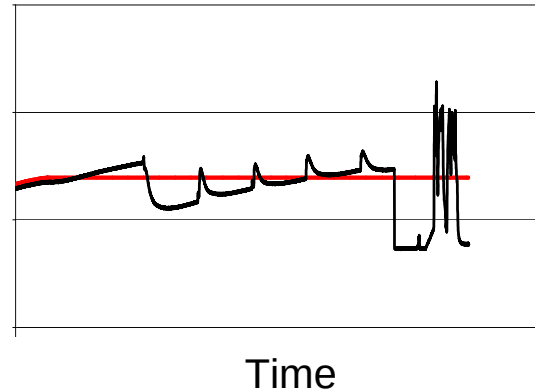


Types of Sensor Failure Modes Targeted

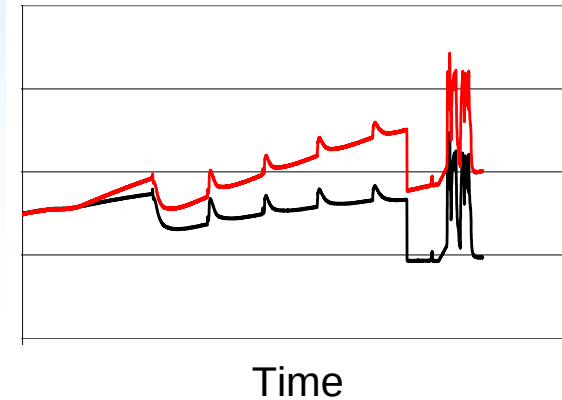
Hard Failure
(open/short circuit)



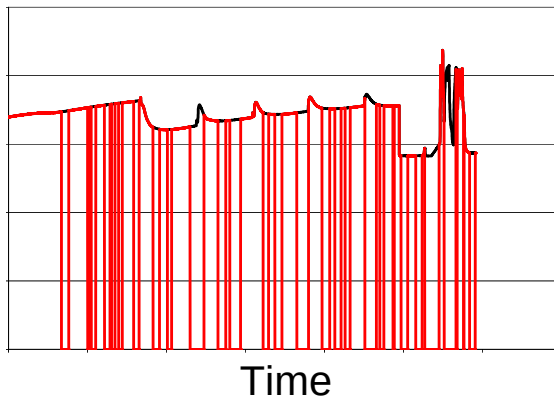
Iced-Up Failure
(sensor value locked)



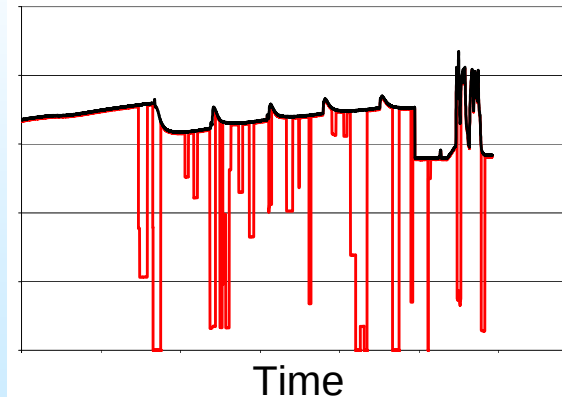
Drift Failure
(thermal/resistance change)



Intermittent-Binary
(loose connector)



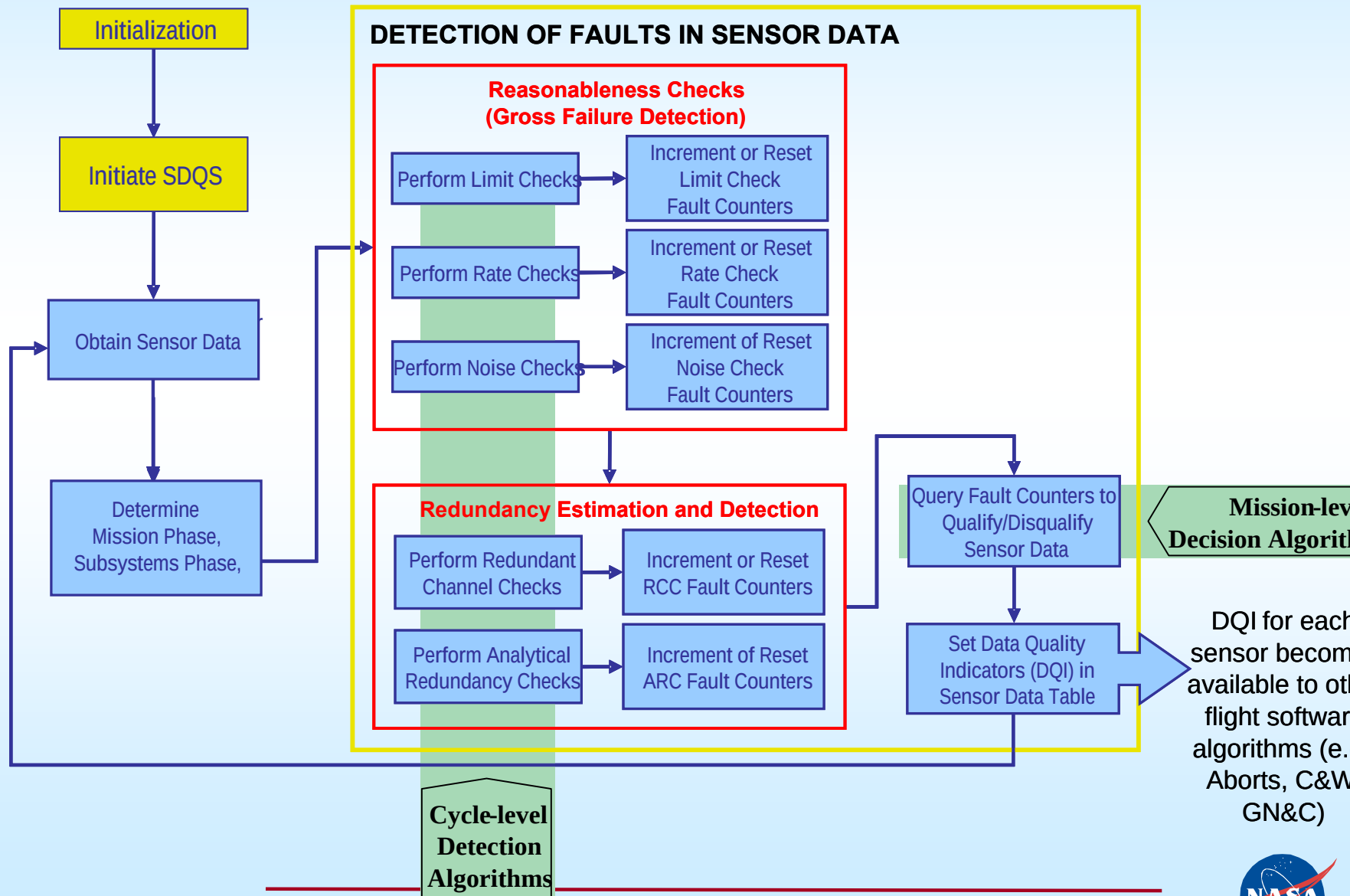
Intermittent-Filtered
(cracked solder joint)



Legend

- Good Sensor
- Failed Sensor

Sensor Data Qualification System (SDQS) Functional Flow



Portable Health Algorithms Test (PHALT) System

The **PHALT** System was developed for use in rapid prototyping and testing of diagnostic algorithms in real-time hardware

- ◆ **Portable**

Laptop (development platform) and industrial, rack-mount PC (real-time target) provide portability to support on-the-road demonstrations and real-time testing

- ◆ **Diagnostic Algorithms**

- ◆ Currently limited to data validation
- ◆ Capability to add a variety of diagnostic & prognostic health management algorithms

- ◆ **Test**

- ◆ Matlab/Simulink xPC software provides capability for rapid prototyping and seamless generation of real-time applications.
- ◆ Industrial PC with real-time I/O supports real-time testing of algorithms with broad spectrum of available test rigs.

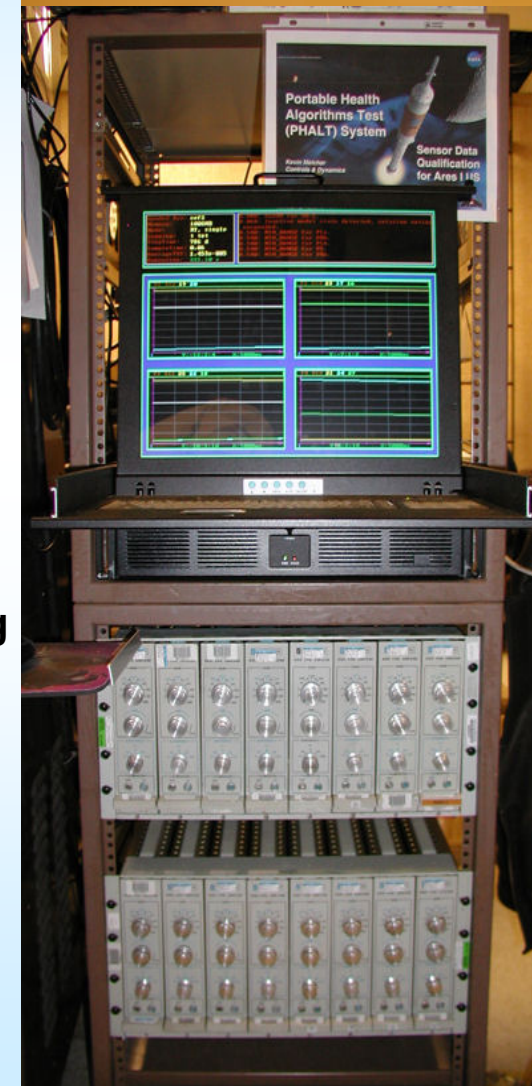
Real-time
Target with
GUI & I/O



Anti-aliasing
Filters



PHALT System



PHALT System Demonstration Results

Data Qualification Testbed Demonstrations

- PHALT System tied onto the Electrical Power System (EPS) Power Distribution Unit (PDU) Controller Area Network (CAN) Bus Card
 - 25 of 26 Simulated Sensor failures were caught by the qualification network being demonstrated.
 - The Undetected Failure Was a Drift That Would Have Been Caught if Allowed to Continue.
- PHALT System implemented within a Combustion Valve Test Rig
 - Real-time hardware sensor failures produced
 - Conducted demonstration of sensor failures with closed-loop control sensors
 - Demonstrated viability of software approach
 - Tests and evaluate of detection software are on-going

TRANSITION OF SENSOR SYSTEMS TO FLIGHT: SOME AREAS OF CONSIDERATION

- **DEVELOPMENT OF A FULL LIFE-CYCLE PLAN FOR THE PRODUCT IS FUNDAMENTAL**
 - **TEAM COMPOSED OF THE PRODUCT DEVELOPER, THE END-USER, AND FLIGHT VEHICLE AND/OR GROUND SUPPORT . EXPERTS FROM SAFETY, RELIABILITY, LOGISTICS, SYSTEM INTEGRATION AND PROJECT MANAGEMENT.**
- **DETAILED SET OF REQUIREMENTS CRITICAL VERY EARLY IN THE PROCESS TO AVOID UNNECESSARY DELAYS AND/OR COSTLY REDESIGNS. INCLUDED ARE:**
 - **PERFORMANCE CONSIDERATIONS**
 - **PHYSICAL CONSIDERATIONS**
 - **ENVIRONMENTAL CONSIDERATIONS**
 - **SAFETY AND RELIABILITY CONSIDERATIONS**
- **FAILURE MODES EVALUATION ANALYSIS CONDUCTED DURING THE QUALIFICATION OF THE PRODUCT.**
- **EXTENSIVE TESTING EVEN TO FAILURE**

SYSTEM LEVEL SUGGESTIONS

- **SENSORS SYSTEMS BE INCLUDED INTO THE VEHICLE IN THE DESIGN PHASE.**
- **STUDY THE VEHICLE SYSTEM TO DETERMINE OPERATIONAL FUNCTION AND CRITICALITY OF VARIOUS SENSOR SYSTEMS AND HOW TO OPTIMIZE CROSS FUNCTIONALITIES.**
- **INSTRUMENT THE VEHICLE SYSTEM TO ALLOW MEASUREMENTS TO ENABLE DAMAGE/DEGRADATION PREDICTION AT A LEVEL TO ALLOW AUTONOMOUS OPERATION.**
- **DEMONSTRATE SENSOR RELIABILITY AND DURABILITY BEFORE INCLUSION OF SENSOR SYSTEM INTO VEHICLE.**
- **PERFORM SENSOR MEASUREMENTS TO OPTIMIZE MEASUREMENT OF MULTIPLE PARAMETERS SIMULTANEOUSLY TO IMPROVE FULL-FIELD SYSTEM INFORMATION AND MEASUREMENT RELIABILITY**
- **DEVELOP SENSOR SYSTEMS WHICH INCLUDE INTEGRATED INTELLIGENCE WHILE MINIMIZING SIZE, WEIGHT, AND POWER CONSUMPTION.**
- **AT MINIMUM, CRIT 1 SYSTEMS, I.E. THOSE WHOSE FUNCTION CAN AFFECT LOSS OF CREW AND/OR VEHICLE, SHOULD BE MONITORED NO MATTER THE EXTREME CONDITION INHERENT IN SUCH MONITORING**

FUTURE DIRECTIONS

NANOTECHNOLOGY

NANOTECHNOLOGY DEVELOPMENT

NANO DIMENSIONAL CONTROL PREVALENT IN CHEM/BIO SENSORS

- **NANO CONTROL OF CHEMICAL SENSOR STRUCTURES STRONGLY PREFERRED EVEN IF SENSOR ISN'T LABELED A "NANO SENSOR"**
 - **WE ARE MEASURING VARYING NUMBERS OF MOLECULES**
- **IF NANOTECHNOLOGY ALREADY PRESENT IN CHEM/BIO SENSOR DEVELOPMENT, THEN:**
 - **WHAT STAYS THE SAME AND WHAT'S NEW?**
 - **WHAT ARE THE CHALLENGES IN NANOTECHNOLOGY DEVELOPMENT?**
 - **WHAT IS THE ROLE/ADVANTAGE OF NANO TECHNOLOGY**

SAME

- **APPLICATIONS DON'T CARE THAT IT IS NANO, NEED IMPROVED CAPABILITIES**
- **STANDARD SENSOR TECHNOLOGY REQUIREMENTS, POTENTIAL, AND DIRECTIONS SET BY THE ADVENT OF MICRITECHNOLOGY REMAIN CONSTANT**
- **SENSITIVITY, SELECTIVITY, STABILITY, RESPONSE TIME, TAILOR FOR THE APPLICATION, "LICK AND STICK", ETC.**
- **PACKAGING STILL SIGNIFICANT COMPONENT OF SYSTEM**
- **AS WITH MICRO, CAN ONLY GO AS FAR AS THE SUPPORTING TECHNOLOGIES**
- **MULTIPLE SENSOR PLATFORMS MAY STILL BE NECESSARY DEPENDING ON THE APPLICATION/ENVIRONMENT**

TARGETED TECHNOLOGY DEVELOPMENT

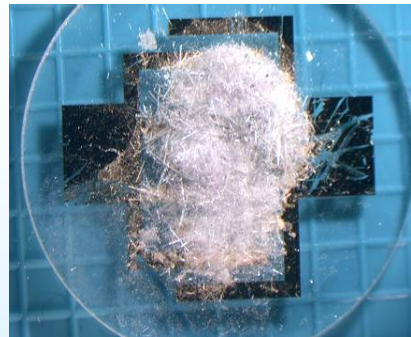
- **MICRO-NANO CONTACT FORMATION**
- **NANOMATERIAL STRUCTURE CONTROL**
- **OTHER NANO OXIDE MATERIALS**

EXAMPLE NANOTECHNOLOGY CHALLENGE: MICRO-NANO CONTACT FORMATION

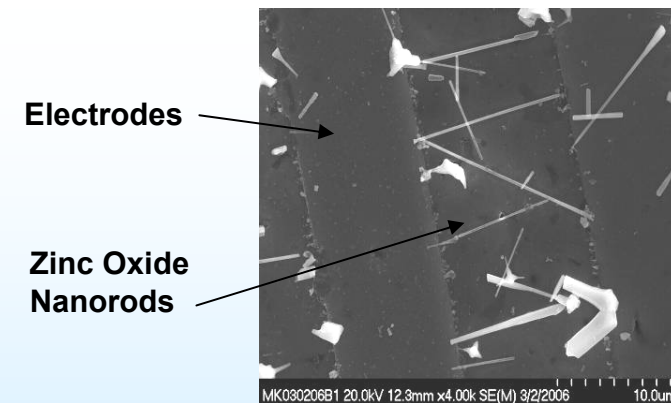
- NO MATTER HOW GOOD THE SENSOR, IF YOU CANNOT MAKE CONTACT WITH IT, THEN IT WILL NOT BE INEFFECTIVE
- MICRO-NANO INTEGRATION/CONTACTS
 - MAJOR QUESTION FOR NANOSTRUCTURED BASED SENSORS: HOW ARE THE NANOSTRUCTURED MATERIALS INTEGRATED INTO A MICROSTRUCTURES
- MANUAL METHODS GENERALLY INVOLVE REPEATABILITY ISSUES E.G.
- BASIC WORK ON-GOING TO IMPROVE MICRO-NANO CONTACTS E.G. USE OF DIELECTROPHORESIS TO ALIGN NANOSTRUCTURES
- BRING THE LEVEL OF PROCESS CONTROL PRESENT IN MICROSYSTEMS TO NANOTECHNOLOGY



**NANOSTRUCTURE
FABRICATED BY THERMAL
EVAPORATION-
CONDENSATION PROCESS.**



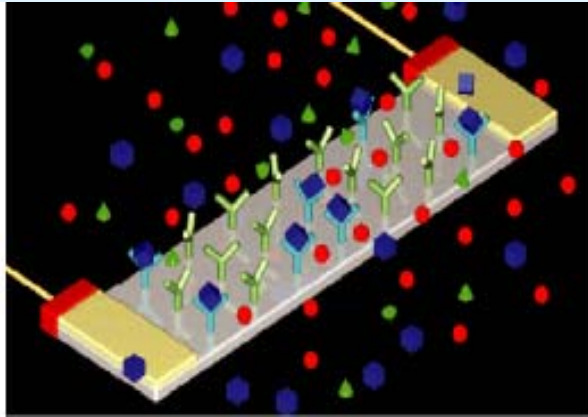
**NANORODS CONTACTED
WITH THE SUBSTRATE VIA
A SILVER EPOXY**



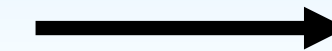
**ZINC OXIDE NANORODS AFT
DIELECTROPHORESIS ACRO
INTERDIGITATED FINGER**

“LOCK AND KEY” CHEMICAL SENSORS USING NANOSTRUCTURES

- **OBJECTIVE: DEMONSTRATE THE FUNDAMENTAL ABILITY TO ASSEMBLE THE ULTIMATE “LOCK AND KEY” CHEMICAL SENSOR DETECTION SYSTEM**
- **STATE OF THE ART:**

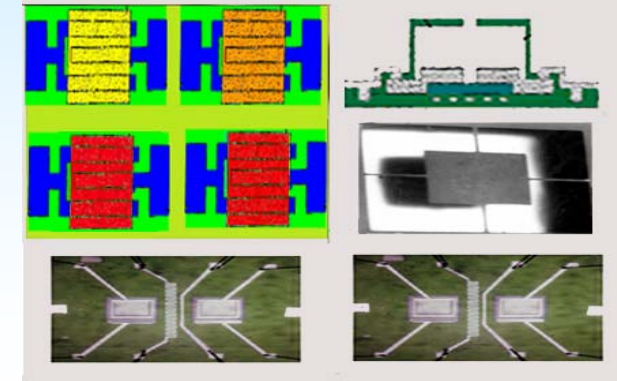


Limited Chemical Selectivity by use
“Lock and Key” Approach



Many Species,
Complex Structures
Lead to Limited
Ability For Species
Identification

NASA GRC HIGH TEMP NOSE APPROACH



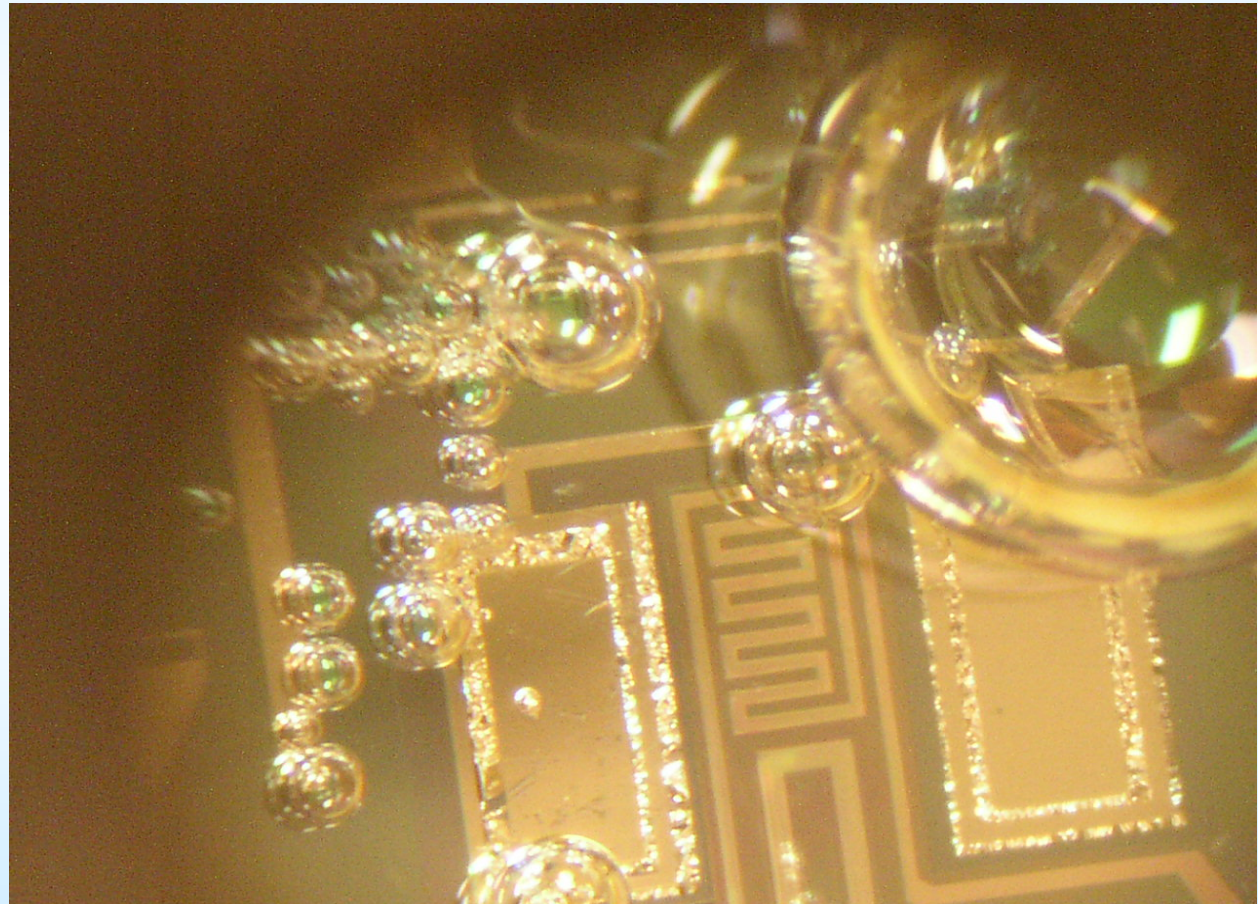
Nose Approaches necessary to attempt to understand
environment but still limited in species identification
multispecies identification, closely related species
significant false positives

- **TECHNICALLY ADDRESS THE FUNDAMENTAL QUESTION “WHAT IS NANO GOOD FOR?”**
- **THE AREA OF CHEMICAL SENSORS:**
 - *NOT* SMALL NANO STRUCTURES FOR BILLION MOLECULE MEASUREMENTS/ IN SUCH APPLICATIONS MAY CONSIDER THIN FILMS OR ALTERNATE SENSOR PLATFORMS
 - *INSTEAD* USE NANOSTRUCTURES FOR DETECTION ON MOLECULAR LEVEL
- **ARRANGE THE CHEMICAL SENSOR STRUCTURE TO “FIT” THE MOLECULE IN QUESTION**
- **VERIFY THE PRESENCE OF THE MOLECULE WITH AN ELECTROCHEMICAL SIGNATURE**
- **FABRICATE “DESIGNER” CHEMICAL SENSORS**

**SENSORS HAVE WIDE VARIETY OF APPLICATIONS BUT CANNOT
WORK IN EVERY ENVIRONMENT**

RIGHT SENSOR FOR RIGHT APPLICATION

H₂ SENSOR OPERATION UNDER WATER



SUMMARY

- **AEROSPACE APPLICATIONS REQUIRE A RANGE OF SENSING TECHNOLOGIES**
- **A RANGE SENSOR AND SENSOR SYSTEM TECHNOLOGY BEING DEVELOPED TO MEET THESE NEEDS USING:**
 - **MICROFABRICATION AND MICROMACHINING TECHNOLOGY**
 - **SMART SENSOR SYSTEMS**
- **TECHNOLOGY BEST APPLIED WITH STRONG INTERACTION WITH USER/TAILORED SENSOR FOR NEEDS OF APPLICATION/SUPPORTING TECHNOLOGIES MANDATORY**
- **SUPPORTING TECHNOLOGIES NECESSARY**
- **DRIVE SYSTEM INTELLIGENCE TO THE SENSOR LEVEL**
- **LONG-TERM: INTELLIGENT SYSTEMS**
 - **RELIABILITY**
 - **REDUNDANCY**
 - **ORTHOGONALITY**
 - **CROSS-CORRELATION**
- **SENSOR IMPLEMENTATION**
 - **A NUMBER OF BARRIERS**
 - **A TEAM APPROACH SUPPORTED BY TESTING**
- **NANOTECHNOLOGY**
 - **SIGNIFICANT PROMISE BUT TECHNOLOGY BARRIERS EXIST**
 - **LONG-TERM FULLY ENABLE “LICK AND STICK”**

ACKNOWLEDGEMENTS

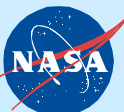
G. Fralick, J. Wrbanek, M. Krasowski, G. Zimmerli, W. Maul, K. Melcher, J. Xu, P. Neudeck, P. Greenberg, G. Fralick, R. Okojie, G. M. Beheim, D. Spry, and L. Matus, NASA Glenn Research Center, Cleveland, OH, USA 44135

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